

Exploring Top Restaurants in Las Vegas: A Data Science Journey!

Background:

As we gear up for our summer trip to Las Vegas, we've decided to delve into the city's vibrant dining scene. To ensure we experience the best and most budget-friendly eateries, we've utilized the [Yelp dataset](#) to identify top-rated restaurants in Las Vegas. We've also filtered out the top 50 fast-food chains to avoid the usual options and embrace more unique culinary experiences.

Project Goal/Stakes:

Our objective is to analyze the Yelp dataset to pinpoint exceptional yet affordable dining establishments in Las Vegas. By focusing on local favorites and hidden gems, we aim to curate a list of restaurants that offer both quality and value, ensuring our trip is both delicious and economical.

Project Submission:

A Note from your Backpacking Partner:

"Hello, team. I've gathered and preprocessed the Yelp dataset for Las Vegas, filtering out the top 50 fast-food chains to ensure we explore the city's unique culinary offerings. Now, it's up to you to analyze this data and find the best budget-friendly restaurants for our trip. I request that you do NOT delete/modify any part of the question description, given code, and pre-loaded comments in the code block.

We look forward to your insightful analyses and recommendations for our culinary adventure in Las Vegas! Hope we can have a wonderful trip in Vegas."

Just a reminder, for a better a smooth autograder experience, please:

- Write your answers between `# ANSWER STARTING HERE` and `# ANSWER ENDS HERE` .
- Do not delete helper cells
- Always use `seed=42` as your `random_state` in the whole notebook.

Submit this Google Colab/Jupyter Notebook file on Gradescope. Cheers! 🍷

Part A (4 Pts Total)

You are expected to load the Yelp data and provide some metrics so I know what we are working with. Follow these steps.

1.1 Import Libraries (0 pt)

Many libraries are imported for you. Import the additional **Sci-kit learn** modules:

- **Decision Tree Classifier**
- **K Neighbors Classifier**
- **Logistic Regression**
- **Random Forest Classifier**

```
In [24]: # Necessary Libraries
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split, cross_val_score, StratifiedKFold
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

import matplotlib.pyplot as plt
import matplotlib
from sklearn.cluster import KMeans, DBSCAN
from sklearn.decomposition import PCA
from sklearn.manifold import TSNE
from sklearn.preprocessing import normalize

from matplotlib.colors import Normalize
from matplotlib.lines import Line2D

import ast
pd.set_option('display.max_columns', None)
from scipy.spatial.distance import cdist

import matplotlib.pyplot as plt
import seaborn as sns
import warnings # To suppress some warnings
import plotly.graph_objects as go
import plotly.express as px

# Suppress the specific FutureWarning
warnings.filterwarnings("ignore", category=FutureWarning, module="seaborn")

# Add the classifiers to the imports here:
# ANSWER STARTING HERE
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier

# ANSWER ENDS HERE
```

1.2 Understanding the Dataset (1 pt)

1. Load the dataset to a dataframe named `yelp_data`.
2. Print the dimensions of the dataset.
3. Print the column names of the dataframe.

```
In [25]: # Load the yelp dataset
# ANSWER STARTING HERE
yelp_data = pd.read_csv("yelp_datasets.csv")
print(yelp_data.shape)
print(yelp_data.columns.tolist)

# ANSWER ENDS HERE

(21318, 16)
<bound method IndexOpsMixin.tolist of Index(['business_id', 'name', 'city', 'state',
'latitude', 'longitude',
      'business_stars', 'review_count', 'categories', 'checkins', 'review_id',
      'user_id', 'reviewer_stars', 'useful', 'funny', 'cool'],
      dtype='object')>
```

```
In [26]: yelp_data.head()
```

```
Out[26]:
```

		business_id	name	city	state	latitude	longitude	business_stars
0	BjH8Xepc10i6OhCDQdX6og	Emeril's New Orleans Fish House	Las Vegas	NV	36.102795	-115.168508	3.	
1	BCAw9a_kyx2RddRH9tMmKQ	Mr. Lucky's 24-7	Las Vegas	NV	36.109100	-115.154526	3.	
2	H8qpFitlesth86zqR4cwYg	Jean Philippe Patisserie	Las Vegas	NV	36.107727	-115.176986	4.	
3	K7IWdNUhCbcnEvI0NhGewg	Wicked Spoon	Las Vegas	NV	36.109538	-115.176170	3.	
4	4mYPOzBokDgz307TW2ElKw	Crafted Buffet	Las Vegas	NV	36.147248	-115.156032	2.	

1.3 Explore the Data (3 pts)

Let's see what is stored in the dataset first to get a feel for what we can expect. Write only a **single line** of code for each.

1. Print the number of **samples** in the dataset (a numerical value).
2. Print the number of **features** in the dataset (a numerical value)
3. **Create two lists of the unique business names and categories**, named `names` and `categories`, respectively. (data type: **list**) Your answer should be similar to:
 - Business: ["Emeril's New Orleans Fish House", "Mr. Lucky's 24-7", 'Jean Philippe' ...]
 - Categories: ['Seafood', 'American (New)', 'Restaurants', 'American (Traditional)', ...]

```
In [27]: # Print the number of samples
# ANSWER STARTING HERE
print("Number of samples:", len(yelp_data) )

# ANSWER ENDS HERE

# Print the number of features
# ANSWER STARTING HERE

print("Number of features:", yelp_data.columns )

# ANSWER ENDS HERE

# Write code to get your two lists.
# ANSWER STARTING HERE

# Extract and collect unique types
names = yelp_data['name'].unique().tolist()

categories = yelp_data['categories'].str.split(';').explode().unique().tolist()

# ANSWER ENDS HERE

print("Business:", names)
print(len(names))
print("Categories:", categories)
print(len(categories))
```

Number of samples: 21318

Number of features: Index(['business_id', 'name', 'city', 'state', 'latitude', 'longitude',
'business_stars', 'review_count', 'categories', 'checkins', 'review_id',
'user_id', 'reviewer_stars', 'useful', 'funny', 'cool'],
dtype='object')

Business: ["Emeril's New Orleans Fish House", "Mr. Lucky's 24-7", "Jean Philippe Pat
isserie", "Wicked Spoon", "Crafted Buffet", "Estiatorio Milos", "Harbor Palace Seafo
od Restaurant", "Gordon Ramsay Steak", "Mon Ami Gabi", "Casa Don Juan - Downtown",
'Round Table Pizza', 'Rose. Rabbit. Lie', 'MGM Grand Hotel', 'Burger Bar', 'Tofu Hu
t', 'Pho Kim Long', 'Soho Japanese Restaurant', "Tina's Filipino Cuisine", 'Noodle S
hop', 'Shake Shack', 'Sushi Kaya', 'The Juice Standard', 'The Buffet', 'Mesa Grill',
'BARDOT Brasserie', 'Hot N Juicy Crawfish', 'Gordon Ramsay Pub & Grill', 'Pho Vietna
mese Cuisine', 'Yunnan Garden', 'Westin Las Vegas Hotel & Spa', 'Chao Thai Restaura
nt', 'Allegro', 'Prommares Thai Food', "Hugo's Cellar", 'Lucki Thai', "Mr D's Sportsb
ar & Grill", 'Buldogis Gourmet Hot Dogs', 'Grand Lux Cafe', "Rosati's Pizza", 'Publi
cUs', 'Tiabi Coffee & Waffle', 'Tacos & Beer', 'Sawa Sushi', 'Paradise Cantina', "Pi
nk's Hot Dogs", 'LAVO Italian Restaurant & Lounge', 'Sweet Poké', "P.F. Chang's", "B
uddy V's Ristorante", 'The Buffet at ARIA', 'Sage', "Gallagher's Steakhouse", 'Littl
e Caesars Pizza', 'America', 'Le Thai', 'The Cheesecake Factory', 'Top of the Worl
d', 'Delhi Indian Cuisine', 'Bacchanal Buffet', "Steak 'n Shake", 'Hooters', 'Jaburr
itos Sushi Burritos', 'Spice Market Buffet', 'T&T - Tacos & Tequila', 'The Bagel Caf
e', 'Le Village Buffet', 'Hash House A Go Go', 'Gangnam Asian BBQ Dining', 'Sushi Ko
ma', 'Los Cucos Mexican Cafe', 'Cabo Wabo Cantina', 'Fogo de Chão Brazilian Steakhou
se', 'La Cave Wine and Food Hideaway', 'Herbs & Rye', 'Kimchi Korean Bbq', 'Krung Si
am Thai', 'RA Sushi Bar Restaurant', 'SW Steakhouse', 'Topgolf', 'Bachi Burger', 'Ei
ffel Tower', "Chili's", 'Biscuits Cafe', 'Egg Works', 'Cucina by Wolfgang Puck', "Ma
ma Depalma's Pizza & Bistro", 'Other Mama', 'Those Guys Pies', 'Off The Strip at the
LINQ', 'Hofbräuhaus Las Vegas', 'Hooters Casino Hotel Las Vegas', 'Border Grill', "G
reenberg's Deli", 'Yardbird Southern Table & Bar', 'Dog Haus', 'California Sushi Bur
rito', 'Giada', 'Tao Nightclub', 'India Palace', "Grimaldi's Pizzeria", "Lola's A Lo
uisiana Kitchen", "Capo's Italian Cuisine", 'Hearthstone Kitchen & Cellar', 'MTO Caf
é', 'Red Lantern Pho Vietnamese Grill', "Maggiano's Little Italy", 'Egg & I', 'New D
ay Cafe', 'The Goodwich', 'Twin Peaks', 'Zenshin Asian Restaurant', 'Cañonita', 'Chi
na Mama Restaurant', 'Secret Pizza', 'Aureole', 'Charlie Palmer Steak', 'Lotus of Si
am', 'Pampas Churrascaria Brazilian Grille', 'Rise & Shine - A Steak & Egg Place',
'Pink Taco', "Andrea's", "Capriotti's Sandwich Shop", "Roberto's Taco Shop", "Marily
n's Cafe", 'Cafe Rio', 'Downtown Terrace', 'Yard House', 'Therapy', 'Chayo Mexican K
itchen + Tequila Bar', 'Morimoto', 'Thai House', 'Luna Lounge', "Guy Fieri's Vegas K
itchen & Bar", 'Babystacks Cafe', "Carrabba's Italian Grill", "Pancho's", 'Bruxie',
'Sumo Sushi 2', 'King & I', 'Blondies Sports Bar & Grill', 'Einstein Bros', 'Fukumim
i Ramen', 'New Grand China', 'Patisserie Manon', 'Todd English P.U.B.', 'Earl of San
dwich', 'Cornish Pasty', 'Flavors The Buffet', "Rocco's NY Pizzeria", 'Kabuki Japane
se Restaurant', 'PBR Rock Bar & Grill', 'Michael Mina Pub 1842', 'Hakkasan Nightclu
b', 'Chubby Cattle', 'Ellis Island Hotel, Casino & Brewery', 'Sushi Bar Sage', 'The
Bootlegger Italian Bistro', 'India Masala Bar & Grill', 'Firefly', 'Joyful House Chi
nese Cuisine', 'Bowlology', 'Hikari', 'Omoide Noodles & Bowls', 'Coronado Cafe', "Ba
ttista's Hole In the Wall", 'China Poblano', 'Holsteins Shakes and Buns', 'Yong Kang
Street', 'Musashi Japanese Steakhouse', 'Lemongrass', 'Julian Serrano Tapas', 'Ameri
can Coney Island Chili Dogs', 'Triple George Grill', 'Pho Kinh Do', 'Costa di Mare',
'Sake Rok', 'The Halal Guys', 'Joël Robuchon', 'Bayside Buffet at Mandalay Bay', 'H
oney Pig', "Robert Irvine's Public House", 'Viet Noodle Bar', 'Pure Indian Cuisine',
'Hard Rock Cafe', 'Fukuburger Truck', "Mount Everest India's Cuisine", 'Hobak Korean
BBQ', 'Public House', "Pop's Philly Steaks", 'Ramen Sora', 'HEXX kitchen + bar', "De
l Frisco's Double Eagle Steak House", "L'Atelier de Joël Robuchon", 'The Peppermill
Restaurant & Fireside Lounge', 'Nobu', 'Sweets Raku', 'Echo & Rig', 'Kabob N More',

'Brio Tuscan Grille', 'Michoacan Gourmet Mexican Restaurant', 'Urban Turban', 'Prime', 'Gordon Biersch Brewery Restaurant', 'The Habit Burger Grill', 'The Buffet at Bellagio', 'Pizza Rock', 'Joes Seafood Prime Steak & Stone Crab', 'Pancho's Vegan Taco s', 'Icy Juicy', 'Teriyaki Madness', 'Olives', 'Paina Cafe', 'Sammy's Woodfired Pizz a and Grill', 'Della's Kitchen', 'Lucille's Smokehouse Bar-B-Que', 'Fleur by Hubert Keller', 'Oscar's Steakhouse', 'Farmer Boys', 'STK Las Vegas', 'Pho Thanh Huong', 'L oftti Cafe', 'El Dorado Express', 'Inaka Sushi', 'PT's', 'TAO Asian Bistro', 'Blue R ibbon Fried Chicken - Downtown Summerlin', 'Bouchon at the Venezia Tower', 'House of Blues - Music Venue', 'Nine Fine Irishmen', 'Chin Chin', 'The Sparklings', 'Scarpett a', 'Kabob Express', 'Blueberry Hill Family Restaurant', 'MadHouse Coffee', 'Brew Te a Bar', 'Pot Tea Social House', 'Metro Pizza', 'Oyshi Sushi', 'Dick's Last Resort', 'La Belle Terre Bread French Bakery Café', 'L & L Hawaiian Barbecue', 'Carmine's Ita lian Restaurant - Las Vegas', 'Market Street Cafe', 'The Smashed Pig Gastropub', 'Fl ippin' Good Chicken, Burgers, Beer', 'Las Vegas North Premium Outlets', 'Archi's Tha i Bistro', 'Red Square Restaurant & Vodka Lounge', 'Sushi Twister', 'Lakeside', 'PKW Y Tavern', 'Strip House Steak House', 'Blue Ox East Restaurant & Lounge', 'Baja Migu el's', 'Lee's Korean BBQ', 'Tacos El Gordo', 'Frank & Fina's Cocina', 'Greens and Pr oteins', 'Senor Frog's', 'Illumilatte Brew Society', 'Slice of Vegas', 'Flour & Barl ey', 'FIX', 'Ocean One Bar and Grille', 'Buffalo Wild Wings', 'Fremont Hotel & Casin o', 'Sunrise Coffee', 'Golden Steer Steakhouse', 'Foundation Room', 'Kyara Japanese Tapas', 'Grouchy John's Coffee', 'Coco's Bakery Restaurant', 'Del Mar Deli', 'Donut Bar', 'Cypress Street Marketplace', 'Monta Ramen', 'Cravings Buffet', 'Yojié Japanes e Fondue', 'Blizz', 'Marché Bacchus', 'Steiner's A Nevada Style Pub', 'Udon Monzo', 'Tony Roma's', 'Liquid Pool Lounge', 'SUSHISAMBA - Las Vegas', 'Double Helix Wine & Whiskey Lounge', 'The Buffet at Luxor', 'Lupo by Wolfgang Puck', 'Thai Style Noodle House', 'Burger Brasserie', 'Beer Park', 'District One', 'Seafood Shack', 'Slidin Th ru', 'Pho Aimie', 'Tonkatsu Kiyoshi', 'Geisha House Steak & Sushi', 'Rise & Shine a Steak & Egg Place', 'Pho Bosa', 'Park on Fremont', 'The Protein Source', 'Lulu's Bre ad & Breakfast', 'Krispy Kreme Doughnuts', 'Bite Breakfast & Lunch', 'Cafe Bellagi o', 'Ruth's Chris Steak House', 'Tony's Mexican Food', 'Aloha Specialties', 'Mr Mama s', 'Pot Liquor CAS', 'Lago', 'Go Vegan Cafe', 'Arawan Thai Bistro and Dessert', 'Sa tay Thai Bistro & Bar', 'Nacho Daddy Downtown', 'Spago by Wolfgang Puck', 'Lawry's T he Prime Rib', 'Teriyaki Boy Healthy Grill', 'Ferraro's Italian Restaurant & Wine Ba r', 'Hyde Bellagio', 'Cleo', 'Du-Pars Restaurant and Bakery', '48th and Crepe', 'Sta ck Restaurant & Bar', 'Broadway Pizzeria', 'The Capital Grille', 'Freed's Bakery', 'Carnevino', 'The Cracked Egg', 'To See Roll', 'Ciao Pizzeria', 'Daisho Sushi', 'La Comida', 'Omelet House', 'Table 10', 'Bouchon Bakery at the Venetian Casino', 'The D iner at LVM Resort', 'Izakaya Go', 'Veggie House', 'ARIA Café', 'Mastro's Ocean Clu b', 'Naked City Pizza Shop', 'Island Flavor', 'Jenni Pho', 'Momofuku Las Vegas', 'Bo cho Sushi', 'Yama Sushi', 'Sushi Way', 'Noodles', 'Tacos N' Ritas', 'Wazuzu', 'Three Angry Wives', 'Lola's A Louisiana Kitchen', 'Hue Thai', 'St. Tropez International Buf fet', 'The Den LV', 'CUT by Wolfgang Puck', 'Cook On Wok', 'Is Sweet', 'Javier's', 'Il Mulino New York', 'Pier 215', 'é by José Andrés', 'Circus Buffet', 'Jean Georges Steakhouse', 'Thai Food To Go', 'Sen of Japan', 'Sura BBQ Buffet', 'Soyo Korean Bars taurant', 'Yellowtail', 'Habaneros Taco Grill', 'Rivas Mexican Grill #2', 'Honey Sal t', 'B&B Ristorante', 'Golden Flower Chinese Cuisine', 'Juan's Flaming Fajitas & Can tina', 'On The Border Mexican Grill & Cantina', 'Four Seasons Diner', 'Voodoo Wing C ompany', '8th Ave Pizza', 'Lazy Dog Restaurant & Bar', 'McCall's Heartland Grill', 'Americana', 'The Pizza Kitchen', 'Taco Tijuana', 'Sultan's Grill', 'Karaoke Q Studi o', 'Makers & Finders Coffee', 'Jason's Deli', 'Gandhi India's Cuisine', 'TENDER Ste ak & Seafood', 'Trattoria Nakamura-Ya', 'Tropical Smoothie Cafe', 'Chicago Joe's', 'The Great Greek', 'Biwon Korean BBQ & Sushi Restaurant', 'Heidi's Brooklyn Deli', 'Noodle Pot', 'Rx Boiler Room', 'Tasty Crepes', 'Vic & Anthony's Steakhouse', 'Carso n Kitchen', 'Cinnaholic', 'The Melting Pot - Las Vegas', 'Dimas Tacos', 'Japanese Cu rry Zen', '808 Sushi', 'Delmonico Steakhouse', 'Hedary's Mediterranean Restaurant', 'Jaleo', 'Ohana Hawaiian BBQ', 'Italian American Club Restaurant', 'Flamingo Pizza',

'Red Robin Gourmet Burgers', 'Aloha Kitchen', 'Margaritaville', 'Lobster ME', 'China Gourmet', 'Jasmine', 'Kumi Japanese Restaurant + Bar', 'Eat.', 'The Olive Mediterranean Grill', 'Culinary Dropout', 'Strip N Dip Chicken Strips', 'BabyStacks Cafe', 'Red Ribbon Bakeshop', 'Ori'Zaba's Scratch Mexican Grill', 'The Henry', 'Buca di Beppo Italian Restaurant', 'Krayvings', 'Sammy's Beach Bar & Grill', 'Snacks', 'Veranda', 'Leticia's Mexican Cocina', 'Eatt Gourmet Bistro', 'Super Burrito', 'Wing Lei', 'Benihana', 'YOLOS', 'Yanni's Greek Grill', 'Eggslut', 'Baguette Cafe', 'The Bar @ Tropicala & Durango', 'Beijing Noodle No. 9', 'Ramen KoBo', 'Kung Fu Tea', 'Paymon's Mediterranean Cafe & Lounge', 'Tea Station', 'POKE POKU Hawaiian Poke bar', 'Dakao Sandwiches', 'Kapit Bahay', 'Casa Di Amore', 'Fiamma Italian Kitchen', 'Sierra Gold', 'Nacho Daddy', 'Full House BBQ', 'Nittaya's Secret Kitchen', 'Sababa Mediterranean Grill', 'Cheffini's Hot Dogs', 'Ohjah Japanese Steakhouse Sushi & Hibachi Flamingo', 'The Lodge Coffee House & Tavern', 'Hokaido Sushi', 'Tamba', 'Va Bene Caffè', 'Cicis', 'Carnegie', 'Mariscos Playa Escondida', 'Viva Las Arepas', 'Mint Indian Bistro', 'The Noodle Man', 'Yassou', 'Surang's Thai Kitchen', 'Downtown Grand Hotel & Casino, An Ascend Hotel Collection Member', 'Downtown Container Park', 'SkinnyFATS', 'Sasa Sushi', 'Bob Taylor's Ranch House', 'Di Fara Pizza', 'India Masala', 'Fatburger', 'Krazy Sushi', 'Wahoo's Fish Tacos', 'Gold Spike', 'Jin Mee Korean BBQ Restaurant', 'The Barrymore', 'Amore Taste of Chicago', 'Wolfgang Puck Bar & Grill Las Vegas', 'Jjanga Steak & Sushi', 'La Creperie', 'Trattoria Reggiano', 'Sambalatte Torrefazione', 'Sushi Roku', 'Florida Cafe Cuban Bar & Grill', 'Bobby's Burger Palace', 'Zizzy', 'Q Bistro', 'Tom's Urban', 'SEA: The Thai Experience', 'Mr Chow', 'VegeNation', 'Casa Del Matador', 'Hennessey's Tavern', 'Nevada Chicken Cafe', 'Northside Café and Chinese Kitchen', 'Brooklyn Bowl', 'Johnny Rockets', 'Pizzeria Francesco's', 'KoMex Fusion', 'Public School 702', 'The Red Pot', 'Stake Out Bar & Grill', 'Crown & Anchor British Pub', 'Olive Garden Italian Restaurant', 'Rí Rá Irish Pub', 'Lindo Michoacan', 'Roxey's Diner', 'Ohjah Japanese Steakhouse Sushi & Hibachi', 'Memphis Championship Barbecue', 'Flock & Fowl', 'Pho Little Saigon', 'Brooksy's Bar & Grill', 'Chica', 'Le Cirque', 'Red 8', 'Rebel Republic', 'bin 702', 'Hussong's Cantina Las Vegas', 'Distill - A Local Bar', 'Grape Street Cafe & Wine Bar', 'Sekushi Japanese Restaurant', 'Casa Don Juan - Summerlin', 'Morels French Steakhouse & Bistro', 'Dom DeMarco's Pizzeria & Bar', 'Cafe Gelato', 'New York Pizza & Pasta', 'Weera Thai Restaurant', 'Ramen Hashi', 'Komol Restaurant', 'SuperBook Deli', 'Gorilla Sushi', 'New York Chinese Restaurant', 'The Buffet At TI', 'Pan Asian', 'Arizona Charlie's Decatur', 'Sushi Loca', 'Jamms Restaurant', 'Settebello Pizzeria Napoletana', 'Yogi's Teriyaki', 'Mr Sandwich', 'Raku', 'Gordon Ramsay Burger', 'Kabuto', 'TangTangTang', 'Bottiglia Cucina & Enoteca', 'Palio', 'TC's Rib Crib', 'Buffet Asia', 'Seasons Buffet', 'Biaggio's Pizzeria', 'Red Sushi', 'Naked City Pizza On Paradise', 'Ito'y Sariling Atin', 'Sam Woo BBQ Restaurant', 'barMASA', 'Market Cafe Vdara', 'Toby Keith's I Love This Bar & Grill', 'Fú Asian Kitchen', 'SK Seafood Restaurant 勝記', 'Bonito Michoacan', 'Kung Fu Thai & Chinese Restaurant', 'Espressamente illy', 'Cheffinis Hot Dogs- THE CART', 'Freedom Beat', 'The Cuppa', 'Twist by Pierre Gagnaire', 'MGM Grand Buffet', 'Broadway Burger Bar & Grill', 'Garden Buffet', 'Trevi', 'Golden Nugget Buffet', 'Fat Dumpling', 'Don Vito's Italian', 'Novecento Pizzeria', 'Xo Chinese Food', 'Ding's Garden', 'Lazeez', 'PizzaRev', 'Rollin' Smoke BBQ', 'La Salsa', 'Gilley's', 'Le Cafe Ile St. Louis', 'Red Rock Pizza', 'Greenland Supermarket', 'Makai Pacific Island Grill', 'Curry Zen', 'Blue Sea Sushi and Izakaya', 'Magnolia's Veranda', 'Silver Sevens Hotel and Casino', 'Social Life Pizza', 'Jose Cuervo Tequileria', 'Seafood City Supermarket', 'Napoli Pizzeria', 'M&M Soul Food Cafe', '800 Degrees Pizzeria', 'Nook Café', 'Violet Bistro', 'Bacon Bar', 'Guy Savoy', 'Magal BBQ', 'Pantry - 24 Hour Kitchen', 'Sporting Life Bar', 'Rustica Pizzeria', 'The Hummus Factory', 'Sicily's Pizza', 'Phil's Italian Steak House', 'Nathan's Famous', 'Black Bear Diner', 'Brewskes Bar & Grille', 'Fleming's Prime Steakhouse & Wine Bar', 'China A GoGo', 'Big Sur Oyster Bar', 'Sugar Factory', 'Asian Wok', 'Albo Pizza', 'J. Gumbo's', 'Pastaidea', 'Sushi Bomb', 'Gyro Time Restaurant', 'Vince Neil's Tatuado EAT DRINK PARTY', 'Camelot Steakhouse', 'Primarily Prime Rib Restaurant', 'Ginseng Korean BBQ II', 'Ramen Ya Katana', 'I

ndia Oven', 'Kona Grill', 'Crunch Donut Factory', 'Robertos Taco Shop', 'Leoné Caf é', 'Paradise Buffet & Cafe Restaurant', 'Lip Smacking Foodie Tours', 'Bahama Breeze', 'WuHu Noodle', 'McFadden's Restaurant and Saloon', 'Dirt Dog', 'Citizens Kitchen and Bar', 'Island Style Restaurant', 'Hussong's Cantina Boca Park', 'Jollibee', 'Pop pa Naps BBQ', 'Cous Cous Mediterranean Cafe', 'Fat Choy', 'Paradise Garden Buffet', 'Backstage Deli', 'Mad House Coffee', 'Mi Casa Grill Cantina', 'Archi's Thai Cafe', 'Pancho's Kitchen', 'Panna Thai Restaurant', 'Tommy Bahama Restaurant | Bar | Store - Las Vegas', 'The Palm Las Vegas', 'Taco Y Taco', 'Streets of New York', 'Joe's New York Pizza', 'Umami Burger - SLS Las Vegas', 'Blue Iguana Mexican Express', 'Gordon Ramsay Fish & Chips', 'Great Links Brewhouse & Grill', 'ENVY The Steakhouse', 'Noodle Asia', 'Texas de Brazil', 'Los Tacos', 'John Cutter', 'Fin', 'Cafe Mayakovsky', 'Tasty Grill', 'Mochiko Chicken', 'StripSteak', 'Viva Mercado's Mexican Bar & Grill', 'Sagos Baja Tavern & Lounge', 'Badger Cafe', 'Ohjah Noodle House', 'Urban Crawfish Station', 'International Eatery', 'Sidestreet Pho and Grill', 'Lazy Joe's Fish & Chips', 'Canaletto', 'Red Rice', 'Sweet Tomatoes', 'Pin-Up Pizza', 'Paradise Place Jamaican Cuisine', 'Sushi Mon', 'Portofino by Chef Michael LaPlaca', 'Ramen Tatsu', 'The Edge of Town', 'Papa Pinny's Pizzeria', 'Wienerschnitzel', 'Don Tortaco Mexican Grill', 'Blaze Fast-Fire'd Pizza', 'Potato Corner', 'GO!KU J-Wraps & Ramen GOKU', 'PDQ - Las Vegas: Sahara', 'Little Crown & Anchor Pub', 'Lucky Streak at LAS', 'Perk Up Coffee Shop', 'Spaghetty Western', 'Coffee Pub', 'China One', 'Harrie's Bagelmania', 'Ichiza', 'NYP- Pizzeria', 'Griddlecakes', 'Paymons Mediterranean Cafe & Hookah Lounge', 'Rachel's Kitchen', 'Familia Pizzeria', 'Smith & Wollensky', 'California Pizza Kitchen', 'Binion's Cafe', 'KJ Kitchen', 'Las Vegas South Premium Outlets', 'Crab Corner', 'Applebee's Neighborhood Grill & Bar', 'SUGARCANE Raw Bar Grill - Las Vegas', 'Forte European Tapas Bar and Bistro', 'South Beach Cafe & Deli', 'Manhattan Pizza', 'Sakana', 'Hot Dog On A Stick', 'Big Dog's Draft House', 'Vintner Grill', 'Tom Colicchio's Craftsteak', 'Gonzalez Y Gonzalez Restaurant', 'Tea Lounge', 'Pizzalicious', 'Las Cazuelas', 'Sinatra', 'Maria's Mexican Restaurant & Bakery', 'Lillie's Asian Cuisine', 'Sirrico's Pizza', 'Sammy's LA Pastrami & Burgers', 'La Reyna Mexican Restaurant & Mariscos', 'Tilted Kilt Pub & Eatery', 'Mingo Kitchen & Lounge', 'Fulton Street Food Hall', 'Fuku Burger', 'Original Chicken Tender', 'Boulevard Bar & Grille', 'Oh Curry', 'Pinches Tacos', '595 Craft And Kitchen', 'Sun's Thai Food & Jerky', 'Lee's Sandwiches', 'Avenue Cafe', 'Charleys Philly Steaks', 'BJ's Restaurant & Brewhouse', 'Wings Restaurants', 'La Maison de Maggie', 'Siena Italian Authentic Trattoria and Deli', 'Osaka Japanese Bistro', 'Pho Saigon 8', 'Hakkasan Restaurant', 'Haute Doggery', 'Glazier's Food Marketplace', 'Chicago Brewing', 'Jjanga Japanese Restaurant', 'Good Fella Korean Bistro', 'Layla Grill & Hookah', 'Cafe Summer', 'Kaizen Fusion Roll and Sushi', 'Chada Thai & Wine', 'Cafe Belle Madeleine', 'Chicago Brewing Company', 'El Sombrero Mexican Bistro', 'Umami', 'Il Fornaio', 'Center Cut Steakhouse', 'Carlos'n Charlie's', 'Carl's Jr', 'KG BBQ', 'Nanny's', 'John Mull's Meats & Road Kill Grill', 'The Martini', 'Planet Hollywood', 'Yi Mei Champion Taiwan Deli', 'Nekter Juice Bar', 'Windy City Beefs N Pizza', 'Frijoles & Frescas Grilled Tacos', 'Blaze Fast Fired Pizza', 'Timbers - Lake Mead', 'The Market', 'Blue Ox Tavern', 'China A Go Go', 'Siamese Bistro', 'Rika Arepa Express', 'Lima Limon', 'North End Pizza', 'Simply Pure by Chef Stacey Dougan', 'Port of Subs', 'Algoberitos', 'Fulton Fish Frye', 'Waffles Cafe', 'Rounders', 'Herringbone', 'Laurita's Taco Shop', 'SUZUYA Pastries & Crepes', 'Sabatino's Authentic NY Pizzeria', 'Seabreeze Cafe', 'Chronic Tacos', 'Viva Lounge', 'Table Thai Bar & Bistro', 'Terrace Pointe Cafe', 'Saffron Flavors of India', 'Wahlburgers', 'Lakes Lounge', 'Pavevino Italian Grille', 'Meraki Greek Grill', 'Margaritaville Casino', 'El Segundo Sol', 'Pieology Pizzeria', 'Rao's', 'Tournament of Kings', 'Sahara Korean Restaurant', 'Sharky's Woodfired Mexican Grill', 'Cafe Americano', 'Veggy Street', 'Joe Maxx Coffee', 'Crazy Pita Rotisserie & Grill', 'Shawarma House', 'Sterling Spoon Cafe', 'Buffet At Asia', 'Zaytoon Restaurant', 'The Hush Puppy', 'Bombay Indian Cuisine', 'The Kitchen at Atomic', 'Goong Korean BBQ Restaurant', 'Yoshi Burritos', 'Cheba Hut', 'Tea Space', 'Off the Hook', 'Hecho en Vegas', 'Stephano's Greek & Mediterranean Grill', 'The Lodge at Shelbourne', 'Napoli Pizza',

"Dickey's Barbecue Pit", 'Big Wong Restaurant', 'Zeffirino', '17 South Booze & Bites', 'Mark Rich's New York Pizza & Pasta', 'Buca Di Beppo', 'Einstein Bros Bagels', 'Sushi Hiro', 'Ho Ho Ho Chinese', 'Roma Deli II and Wine Shop', 'Grotto Ristorante', 'Village Pub and Cafe', 'Rhythm Kitchen', 'La Bocce', 'Montana Meat Company - Durango', 'Bingo at South Point', 'Marias Taco Shop', 'Burger Lounge', 'Roadrunner Saloon', 'Fat Bar', 'GameWorks', 'Pizza My Dear', 'Pyramid Cafe', 'Little Caesar's', 'Taqueria El Buen Pastor', 'Pellegrini Pizza', 'Rubio's', 'China Star', '50's Diner Omelot House', 'Chef Ching's Kitchen', 'Yayo Taco', 'Swish', 'Beau Monde Bakery', 'Buffet of Buffets', 'Aloha Kitchen & Bar', 'Golden Corral', 'Koi Restaurant', 'Bobo China', 'Gyu-Kaku Japanese BBQ', 'Hong Kong Garden Seafood & BBQ Cafe', 'THE Steak House', 'Pour - Coffeehouse', 'Five50 Pizza Bar', 'T-Bones Chophouse', 'The Grand Wok and Sushi Bar', 'Michael's Gourmet Room', 'Ohlala French Bistro', 'Our Families Country Cafe', 'Beach Café', 'Chopstix', 'Kobe Sushi Bar', 'Popeyes Louisiana Kitchen', 'Handel's Homemade Ice Cream & Yogurt', 'TGI Fridays', 'Checker's', 'Max's Restaurant', 'T-Bird Lounge & Restaurant', 'Wolfgang Puck Bar & Grill Summerlin', 'Via Brasil Steakhouse', 'Thai Cuisine', 'Sbarro', 'Mezzo Bistro and Wine', 'Rainbow's End Natural Foods', 'Shabu Grill', 'Hi Scores - Blue Diamond', 'Baja Fresh', 'J Karaoke Bar', 'Stripburger', 'The Rockhouse', 'Picasso', 'The Buffet at Excalibur', 'Amena Bakery & Deli', 'China AAA', 'Top of Binion's Steakhouse', 'Poukei', 'Pepe's Tacos', 'Pop Up Pizza', '8 Noodle Bar', 'Island Sushi & Grill', 'Chart House', 'Bandito Latin Kitchen & Cantina', 'Steiner's', 'Dougie J's Cafe', 'Original Pancake House', 'Mimosas Gourmet', 'Rikki Tiki Sushi', 'Capital Seafood Restaurant', 'Hawaiian Style Poke', 'Habib's Persian Cuisine', 'Hanabi', 'Rainforest Cafe', 'The Tap House', 'Village Pub & Poker', 'Ruby's Diner', 'Tacos La Mexicana', 'Lombardi's Romagna Mia', 'Boba Hut', 'Joe Philly's Pizza', 'Squeeze In', 'Casanova', 'Krazy Buffet', 'Tacos La Carreta', 'Pho 87', 'Roma Deli 1 & Restaurant', 'Rainbow Kitchen', 'Spa Aquae', 'Pon Ocha Soup', 'Sundance Grill', 'Cousins Maine Lobster - Las Vegas', 'Mariscos El Dorado', 'Oyster Bar', 'Heart Attack Grill', 'Otto Enoteca Pizzeria', 'Grand Café', 'Nosh', 'Mandarin Express', 'HK Seafood Garden', 'Rice & Company', 'La Parrilla', 'Lou's Diner', 'Mom's Teriyaki', 'La Bella Napoli Pizzeria', 'Aria Mediterranean Restaurant', 'Fire Rock Steakhouse', 'Market Grille Cafe', 'Grand Cafe', 'Vegas Best Pizza', 'China Hot Wok', 'Tacos Los Toritos', 'Doña Maria Tamales Restaurant', 'Dae Jang Keum Korean BBQ & Tofu Restaurant', 'Mulligan's Pub', 'Azusa Hookah Lounge & Cafe', 'La Parrilla The Lab', 'Virgil's Real Barbecue - Las Vegas', 'Feast Buffet', 'Jefe's Taco Shop', 'Fireside Restaurant & Tavern', 'BLT Steak', 'The Perq', 'Snow White Cafe', 'Kneaders Bakery & Cafe', 'Mary's Hash House', 'Bonefish Grill', 'Sid's Cafe', 'Canaletto Ristorante Veneto', 'Cardenas Market', 'Montana Meat Company', 'Eclipse Theaters', 'P.T.'s Gold', 'Baladie Café', 'Brothers Italian Bistro', 'Sin City Sushi', 'Salsa Loca Mexican Grill', 'Pearl Wok To Go', 'Home Plate Grill & Bar', 'Blue Martini Lounge', 'Cracker Barrel Old Country Store', 'Protein House', 'Chapalas Mexican Restaurants', 'Jacky Chan', 'Fat Tuesday', 'Katsuya', 'Famous Dave's Bar-B-Que', 'Giordanos - Las Vegas', 'Tuscany Gardens', 'Mob Bar', 'Fat Boy', 'Fresh Kabob', 'Zest - Bistro & Bar', 'Cabo Mexican Restaurant', 'Tacos El Compita', 'Naked Fish's Sushi & Grill', 'Stacks & Yolks', 'Primeburger', 'Polaris Street Cafe', 'Kusina Ni Lorraine', 'Starboard Tack', 'Galanga Thai Eatery', 'The Coffee Shop', 'Miller's Ale House - Las Vegas', 'Rubalcabas Taco Shop', 'LOL Kids Club', 'Chophouse & Brewery', 'Paid In Full', 'Bonanno's New York Pizzeria', 'Sonic', 'Ping Pong Thai Restaurant', 'La Mojarra Loca Grill', 'Thai Room', 'DW Bistro', 'Sushi Culiacan', 'Westcliff Station', 'JJ's Boulangerie', 'Bogey's Bar & Grill', 'Cuba Cafe', 'Jardin', 'Bubba Gump Shrimp', 'Nanay Gloria', 'Wolfgang Puck Express', 'Bambino's East Coast Pizzeria', 'Marketplace Cafe', 'Crêpe Station', 'Ketchup Premium Burger Bar', 'Chip Off The Block Deli', 'Mian Sichuan Noodles', 'H&H BBQ Plus 2', 'MOzen Bistro', 'Martorano's', 'Fausto's Mexican Grill', 'Thai Pan Cuisine', 'No Regrets Bar', 'Barcelona Tapas', 'California Noodle House', 'Muscle Maker Grill - Las Vegas', 'The Black Sheep', 'Sushi Fever', 'Blossom', 'E-Jo Korean Restaurant', 'Scooter's Pub', 'Fleming's Prime Steakhouse', 'Sushi Neko', 'Taco Time', 'Bacio', 'Teppan Bento', 'Bazaar Meat by José Andrés', 'Oasis Asian

Kitchen', 'Hard Hat Lounge', 'Thai Pepper', 'Brown Rice Thai Kitchen', 'Fruits and R
oots', 'Rice n Noodle', 'Claim Jumper', 'Taco Maker', 'Angelina's Pizzeria - Easter
n', 'Piero's Italian Cuisine', 'Fresh Buffet', 'Don Chuy Mexican Grill', 'Yogurt I
n', 'Daily Kitchen Modern Eatery and Rotisserie', 'Little Italy Pizza', 'Frankie's P
izza', 'The Bar @ Las Vegas Blvd & Wigwam', 'Old Homestead Steakhouse', 'IKEA', 'Yum
my Grill & Sushi', 'Cherry Berries', 'Burke in the Box', 'AquaKnox', 'Beauty & Esse
x', 'Shucks Tavern & Oyster Bar', 'Violette's Vegan Organic Cafe & Juice Bar', 'Taco
Bell Cantina', 'Sin Fronteras Tacos', 'Blue Ox Central', 'Island Flavor ', 'Beach Ca
fe', 'Overlook Grill', 'Joey's Tavern', 'Anna Marie's Italian Cuisine', 'Denny's',
'Chun Fai Chinese Eatery', 'North Beach Italian Bistro', 'Tennesseseasonings', 'Kopper
Keg North', 'Outback Steakhouse', 'Luke's Lobster Las Vegas', 'Original Tommy's Hamb
urgers', 'Dave & Buster's', 'Bar Code Burger Bar', 'Rampart Buffet', 'Giuseppe's Bar
& Grille', 'Searsucker', 'Tacos El Compita 3', 'Golden China Chinese Cuisine', 'Mr.
ChopStix', 'Soul Food Cafe', 'Pho T', 'Rock & Reilly's Las Vegas', 'Libertine Socia
l', 'WTPho', 'S7 Buffet', 'Coney Island Gourmet Hotdogs', 'Rilibertos Fresh Mexican
Food', 'Italian Oasis Pizzeria', 'Serrano's Mexican Restaurant', 'Scentz N Sudz', 'N
estle Toll House Cafe by Chip', 'Asmara Restaurant', 'Winchell's Pub & Grill', 'Juic
e 'N' Go', 'Mother's Korean Grill', 'Cafe Breizh', 'Vesta Coffee Roasters', 'San Sal
vador Restaurant', 'Le Petit Café & Bakery', 'Pineapple Park', 'I Am Thai Cuisine',
'Twin Creeks', 'Panini Cafe', 'Zuma Las Vegas', 'Timbers - Novat', 'Pamplemousse Le
Restaurant', 'The Rice Shop', 'San Gennaro Feast', '1923 Bourbon & Burlesque', 'Dea
n's Place', 'Boca Do Brasil', 'Timbers - Durango', 'Delices Gourmands French Baker
y', 'Mastrioni's', 'Basil 'n Lime', 'Vegeway', 'The Original Sunrise Cafe', 'Las Pup
usas', 'Blue Ribbon Fried Chicken - Grand Bazaar', 'Michael Mina', 'Abuela's Tacos',
'Red Lobster', 'Mimi's Cafe', 'Cafe Zupas', 'Hawthorn Grill', 'Le Pho Vietnamese Kit
chen', 'Charlie's Bar + Grill', 'Ronald's Donuts', 'Lilly's Thai & Vietnamese Restau
rant', 'Villa Pizza Durango', 'Sammy's Woodfired Pizza And Grill', 'Marie Callende
r's Restaurant & Bakery', 'Buffet Asia', 'Ichi Ramen House', 'Pizza Forte', 'Thai Or
iginal BBQ', 'Carmine's Pizza Kitchen', 'Big Ern's BBQ', 'Golden China', 'Tacos Mexi
co', 'D Grill', 'Mr Poké Bowls & Rolls', 'Rick's Rollin Smoke BBQ & Tavern', 'Ichabo
d's Lounge', 'Easy Life Boba Tea', 'Mizuya', 'Metro Diner', 'Fist of Fusion Island G
rill', 'Fuddruckers', 'Above the Crust', 'Silverado Steakhouse', 'Rosallie', 'Libre
Mexican Cantina', 'Sterling Brunch', 'Divine Eatery', 'SG Bar', 'Marrakech', 'Cante
r's Las Vegas', 'Oyster Bay Seafood', 'Frijoles & Frescas', 'Bjs Cocktail Lounge Dec
atur', 'Design & Dine', 'MAMAOH', 'Mizumi', 'Rocky's Bar and Grill', 'Crush', 'Hong
Kong Garden Seafood & Dim Sum Cafe', 'Curbside Cafe', 'Hong Kong Cafe', 'Pie Me Ove
r', 'Vitality Bowls', 'Buca di Beppo', 'Tups Specialty Sandwiches', 'Soho Sushi Burr
ito', 'Verrazano Pizza', 'Mr. Z Tea Co', 'The Lodge at Grand Teton', 'Japaneiro', 'J
oe Vicari's Andiamo Italian Steakhouse', 'Porchlight Grille', 'Petrossian Bar', 'Roc
co's', 'Beerhaus', 'Macayo Vegas', 'Fun Tacos', 'Sparrow + Wolf', 'Great Wall Chines
e Restaurant', 'Joshua Filipino Oriental Market', 'Luckys', 'The Mixx Grill and Loun
ge', 'Sicilian Ristorante', 'Andiron Steak & Sea', 'Chengdu Taste', 'Arata Sushi & B
owl', 'Chuck E. Cheese's', 'Cugino's Italian Deli & Pizza', 'Parsley Modern Mediterr
anean', 'Plantone's Italian Market', 'Pho 88', 'Sushi-ko', 'Thai Seafood Ginger', 'N
othing Bundt Cakes', 'Las Vegas Chinatown Plaza', 'Rosati's', 'Pei Wei Asian Diner',
'El Diamante Restaurant', 'Rivea', 'Moko Ramen Bar', 'Pho Vegas', 'Turmeric Flavors
of India', 'The Coffee Bean & Tea Leaf', 'Toros Cantina & Gaming', 'Chada Street',
'Rincon Catracho', 'Bavette's Steakhouse & Bar', 'Mickie Finnz', 'The Lodge Hualapa
i', 'The Lodge Tenaya', 'Cevicheria El Diamante', 'Rolling in Dough Cafe', 'Russo's
Pizza Kitchen', 'Tina's Gourmet Sausage House', 'Chili's Too', 'Curry Pot', 'Asian B
BQ & Noodle', '8oz Korean Steak House', 'Philly Steak Express', 'Patty's Tamales',
'Harvest by Roy Ellamar', 'The Valley Thai Restaurant', 'Bonsai Japanese Steak & Sus
hi', 'The Green Regime', 'Mulligan's Border Bar & Grill', 'Mantra Masala', 'Tapioca
Express', 'Ocha Thai Cuisine', 'Drugstore Cafe', 'Pizza Company', 'Port of Subs #11
4', 'Ely's Restaurant', 'Kaiseki Yuzu', '7 Sinful Subs', 'McDonalds', 'Magnolia Ice
Cream and Treats', 'Mr Beijing', 'Sambalatte', 'Apache Taco Shop', 'Grinders Pizza L

ounge', 'Pressed For Juice', 'Rincon De Buenos Aires', 'Presto Calzone Bakery', 'China Dragon', 'Marco's Pizza', 'Siegel's 1941', 'John's Incredible Pizza Company', 'The Sandwich Spot', 'Doh Korean BBQ', 'TAP Sports Bar', 'Lanai Express', 'Cafe Moda', 'Kapuso Kapamilya', 'Tacos Huevos', 'Afandi Restaurant and Market', 'The Great American Bagel', 'Papa John's Pizza', 'Lobster Me', 'Samurai Sam's', 'El Burrito Mexican Restaurant', 'Aloha Hawaiian BBQ', 'Moo Woo Korean BBQ', 'Timbers - Cheyenne', 'Retro Bleu Cafe', 'Sushi Takashi', 'Cafe Sanuki', 'Onion Steak Sub', 'WingNutz', 'Salvatore's', 'Crazy Crepes', 'Villa Fresh Italian Kitchen', 'Canyon Ranch Grill', 'Food Express Chinese Restaurant', 'Tequila Taqueria', 'The Lodge Cactus', 'Portion Control', 'Zabas Mexican Grill at the Plaza', 'DJT', 'Andre's Bistro & Bar', 'Hennessey's Las Vegas', 'Evel Pie', 'Hometown Eats', 'Arirang Korean BBQ', 'Chop Chop Wok', 'Cactus Jacks Saloon', 'Oiga, Mire, Vea Colombian Cuisine', 'Cafe Modena', 'Palm Pizza', 'Costco Food Court', '360 Gourmet Burrito', 'Los Compadres Meat Market', 'Anise Taps & Grill', 'Maestro's Pizzeria', 'Ninja Teriyaki - Sushi 2 Go', 'RM Seafood', 'Bentley's Bar and Grill', 'Don Tortaco's', 'Binion's Gambling Hall', 'Paris Baguette', 'Straight From Philly Steakout', 'Great Harvest Bread', 'Tony Big's Pizza N Pasta', 'Sicili's Pizza', 'Capriotti's', 'Noodle King', 'Veggie Delight', 'Kuben', 'Chuchote Thai Bistro & Desserts', 'Bronze Cafe at The Market', 'CJ's Italian Ice & Custard', 'Plumeria Thai Cafe', 'Chile Verde Express Mexican Grill', 'Rani's World Foods', 'Manna Korean BBQ', 'Savona Bistro', 'Wo Fat', 'Mercato Della Pescheria', 'Lobster Chowder Company', 'Dona Maria Tamales Restaurant', 'The Oyster Bar', 'Ru Yi Noodle House', 'O-my Sushi', 'Grill 55', 'Blue Burrito Grille', 'Magura', 'Great Harvest Bread Company', 'Centro America Restaurant', 'Lino's Pizzeria & Italian Cuisine', 'Tacos Y Tortas El Panzon', 'Farm Basket', 'Sourdough Cafe', 'Caffè Bottega', 'Jack In the Box', 'Bomas Bar & Grill', 'Groundswell Tabletop Game Cafe', 'Pearl Ocean', 'Boston Pizza', 'Burgerim', 'Harvest Cafe', 'Asian Garden', 'Santa Fe Mining Company', 'Song's Family', 'Volcano Grille', '702 Wing Spot', 'Jaburritos', 'Emperor's Garden', 'Pizzeria', 'Shoku Ramen', 'Islands Restaurant', 'Tableau', 'Stir Krazy Mongolian Grill', 'Aloha A Go Go', 'Pizzeria Da Enzo', 'Vegeway', 'Pei Wei', 'Carbone', 'Sushi Tower & Steakhouse', 'Fuzzy's Taco Shop', 'Sushi Hiroyoshi Japanese Cuisine', 'China One - South Durango', 'Time-Out Sports Bar & Grill', 'Strip View Cafe', 'Ohana Hawaiian Bbq', 'Teriyaki House', 'Yanni's Gyros', 'Jackie Bee Boba Express', 'Chopstix Express', 'Famous Dave's Barbeque', 'Yui Edomae Sushi', 'LJ Filipino Restaurant and Pub', 'MrJims.Pizza of Las Vegas', 'Cariba Charlie's', 'Puerto Rico Express', 'JC Cuisine', 'The Pint', 'Pollo Inka Express', 'El Steak Burrito', 'Rocco's Ny Pizza & Pasta', 'Mexicali Raspados', 'Julio's Tacos & Grill', 'Westside Deli', 'Maria's Taco Shop', 'Village Pub & Grill', 'Chow Mein House', 'The Still', 'Grand Bazaar Shops', 'Great Buns Bakery', 'Fiddlestix', 'Pho 90', 'Sofia's Cafe', 'Pho Bistro', 'Hunan Restaurant', 'Esmeralda's Cafe', 'L&L Hawaiian Barbecue', 'Sister's Oriental Market & Video', 'The Chicken Shack', 'Little Dumpling', 'Polish Deli', 'Las Pupusas Restaurant', 'Fiiz Drinks Las Vegas', 'Perú Chicken', 'China a Go Go', 'Happy Wok', 'PkwY Tavern', 'Valerio's Tropical Bake Shop', 'Casa Don Juan Mexican Restaurant', 'Goldfinger's Bar & Grill', 'Trattoria Italia', 'Sofrito Rico Authentic Puerto Rican Cuisine', 'Dirt Dog - Las Vegas Strip', 'Las Famosas de Jose', 'Hachi', 'Tail and Fin', 'Canyon Ranch Cafe', 'MB Steak', 'Yunnan Tasty Garden', 'Rock n Roll Sushi', 'Goldilocks', 'Citrus at the Downtown Grand Pool Deck', 'Pho Vietnam Restaurant', 'Tom Colicchio's Heritage Steak', 'Johnny Vegas Grill', 'Fellini's Ristorante', '7th & Carson', 'Benny's Smokin BBQ and Brews', 'Ciao Ciao Cafe & Bar']

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Categories: ['Seafood', 'American (New)', 'Restaurants', 'American (Traditional)', 'Diners', 'Specialty Food', 'Chocolatiers & Shops', 'Food', 'Desserts', 'Bakeries', 'Buffets', 'Breakfast & Brunch', 'Greek', 'Chinese', 'Cantonese', 'Dim Sum', 'Salad', 'Soup', 'Steakhouses', 'French', 'Mexican', 'Pizza', 'Arts & Entertainment', 'Bars', 'Performing Arts', 'Nightlife', 'Hotels', 'Event Planning & Services', 'Casinos', 'Resorts', 'Hotels & Travel', 'Burgers', 'Korean', 'Barbeque', 'Vietnamese', 'Sushi Bars', 'Asian Fusion', 'Japanese', 'Filipino', 'Thai', 'Szechuan', 'Juice Bars &

Smoothies', 'New Mexican Cuisine', 'Southern', 'Tex-Mex', 'Cafes', 'Brasseries', 'Hotels', 'Cajun/Creole', 'Gastropubs', 'Fish & Chips', 'British', 'Pubs', 'Ice Cream & Frozen Yogurt', 'Noodles', 'Food Delivery Services', 'Italian', 'Local Flavor', 'Sports Bars', 'Karaoke', 'Hot Dogs', 'Chicken Wings', 'Sandwiches', 'Coffee & Tea', 'Internet Cafes', 'Comfort Food', 'Latin American', 'Tapas/Small Plates', 'Cocktail Bars', 'Fast Food', 'Dance Clubs', 'Lounges', 'Hawaiian', 'Poke', 'Gluten-Free', 'Mediterranean', 'Beer Gardens', 'Indian', 'Halal', 'Vegetarian', 'Bagels', 'Delis', 'Caterers', 'Music Venues', 'Brazilian', 'Tapas Bars', 'Active Life', 'Golf', 'Beer', 'Wine & Spirits', 'Live/Raw Food', 'Cheesesteaks', 'Beer Bar', 'Beer Garden', 'German', 'Wine Bars', 'Street Vendors', 'Tacos', 'Taiwanese', 'Car Dealers', 'Automotive', 'Grocery', 'Hookah Bars', 'African', 'Waffles', 'Ramen', 'Wraps', 'Irish', 'Do-It-Yourself Food', 'Hot Pot', 'Fondue', 'Mongolian', 'Breweries', 'Venues & Event Spaces', 'Pakistani', 'Vegan', 'Spanish', 'Acai Bowls', 'Middle Eastern', 'Food Stands', 'Food Trucks', 'Meat Shops', 'Butcher', 'Arabian', 'Tuscan', 'Bubble Tea', 'Smokehouse', 'Shaved Ice', 'Dive Bars', 'Chicken Shop', 'Musicians', 'Persian/Iranian', 'Afghan', 'Kebab', 'Tea Rooms', 'Shopping Centers', 'Fashion', 'Outlet Stores', 'Shopping', 'Russian', 'Local Services', 'Art Galleries', 'Donuts', 'Swimming Pools', 'Ethnic Food', 'Bed & Breakfast', 'Soul Food', 'Organic Stores', 'Fruits & Veggies', 'Malaysian', 'Cupcakes', 'Macarons', 'Creperies', 'Lebanese', 'Personal Chefs', 'Tepanyaki', 'Falafel', 'Kosher', 'Venezuelan', 'Pasta Shops', 'Piano Bars', 'Amusement Parks', 'Cuban', 'Irish Pub', 'Bowling', 'Singaporean', 'Coffeeshops', 'Gelato', 'Shanghaiese', 'Ethical Grocery', 'Seafood Markets', 'Candy Stores', 'Food Tours', 'Tours', 'Walking Tours', 'Laotian', 'Caribbean', 'Guamanian', 'Modern European', 'Ukrainian', 'Patisserie/Cake Shop', 'Japanese Curry', 'Delicatessen', 'Convenience Stores', 'Peruvian', 'Bingo Halls', 'Arcades', 'Portuguese', 'Health Markets', 'Nutritionists', 'Vitamins & Supplements', 'Health & Medical', 'Weight Loss Centers', 'Party & Event Planning', 'Gift Shops', 'Beauty & Spas', 'Hair Salons', 'Soba', 'Flowers & Gifts', 'Massage', 'Day Spas', 'Pan Asian', 'Salvadoran', 'Cinema', 'Dinner Theater', 'Pretzels', 'Playgrounds', 'Kids Activities', 'Scandinavian', 'Furniture Stores', 'Kitchen & Bath', 'Baby Gear & Furniture', 'Home & Garden', 'Home Decor', 'Pool Halls', 'Eatertainment', 'Screen Printing', 'Printing Services', 'South African', 'Personal Shopping', 'Cosmetics & Beauty Supply', 'Ethiopian', 'Coffee Roasteries', 'Theme Cafes', 'Real Estate', 'Mortgage Brokers', 'Festivals', 'Home Services', 'Moroccan', 'Paint & Sip', 'Hong Kong Style Cafe', 'Bangladeshi', 'Books', 'Mags', 'Music & Video', 'Videos & Video Game Rental', 'Honduran', 'Argentine', 'Education', 'Colleges & Universities', 'Colombian', 'Department Stores', 'Food Court', 'Izakaya', 'Tabletop Games', 'Puerto Rican', 'Wholesalers', 'Professional Services', 'Beverage Store', 'Farmers Market', 'Imported Food']

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Part B (9 Pts Total)

Now we need to do some scraping of the dataset so you can feed it into the training model.

2.1 Preprocessing (3 pts)

Before training the model, we need to clean and prepare the data. Please do the following.

1. Print the number of missing data in each column. Please specify which column name you got the number from. You can create a **dictionary** `missing_data`.
2. Impute missing values in the `yelp_data` dataset by filling them with zeros. Save the processed data as `yelp_data`.

```
In [28]: # Print the number of missing data in each column.
# ANSWER STARTING HERE
missing_data = yelp_data.isnull().sum().to_dict()
print(missing_data)

# ANSWER ENDS HERE

# ANSWER STARTING HERE

yelp_data = yelp_data.fillna(0)

# ANSWER ENDS HERE
```

```
{'business_id': 0, 'name': 0, 'city': 0, 'state': 0, 'latitude': 0, 'longitude': 0,
'business_stars': 0, 'review_count': 0, 'categories': 0, 'checkins': 0, 'review_id':
0, 'user_id': 0, 'reviewer_stars': 13, 'useful': 12, 'funny': 13, 'cool': 0}
```

2.2 Splitting, Training, and Testing (6 pts)

Typically in machine learning, we wish to split the data into various portions in order to let a machine learning model train itself on one part of the data and test itself on another part of the data. Splitting also helps prevent overfitting, ensuring the model can make accurate predictions on new, unseen data.

Training data is used to learn the model, while testing data assesses its generalization to unseen examples. I would recommend using **80%** of the data for training and **20%** for testing.

Before we split the data, we are going to help you process the raw data a little bit more:

1. We prefer restaurants close to Canyon Gate in Las Vegas, near our AirBnb.
2. We add a `preference_rating` column at the end of the dataframe, so that we can predict the rating classification.
3. We add non-linearity to `preference_rating`

```
In [29]: # Helper Function, Do Not Remove
import numpy as np
import pandas as pd
import hashlib
from sklearn.preprocessing import MinMaxScaler

def business_id_to_float(business_id):
    hash_object = hashlib.md5(business_id.encode())
    hash_hex = hash_object.hexdigest()
    hash_int = int(hash_hex[:8], 16)
    return hash_int / (2**32 - 1)

def float_to_business_id(unique_float, original_business_ids):
    id_to_float = {id: business_id_to_float(id) for id in original_business_ids}
```

```

closest_id = min(id_to_float.keys(), key=lambda x: abs(id_to_float[x] - unique_
return closest_id

# Non-linear transformation functions
def apply_non_linear_transformation(x, steepness=2):
    """
    Apply a sigmoid-like non-linear transformation
    - Increases sensitivity around mid-range values
    - Compresses extreme values
    """
    return 1 / (1 + np.exp(-steepness * (x - 0.5))) * 8

def exponential_boost(x, base=1.5):
    """
    Apply exponential boosting to favor higher values
    """
    return np.power(x, base)

def angular_score_adjustment(x):
    """
    Use trigonometric function to add non-linear variation
    """
    return np.sin(np.pi * x / 1.5) * 2

# Define weights
weights = {
    'location_score': 0.15,
    'business_stars': 0.35,
    'reviewer_stars': 0.35,
    'engagement_score': 0.15
}

# Define the target location as Canyon Gate, Las Vegas
target_location = (36.1433, -115.2821)

# Haversine distance calculation
def haversine(lat1, lon1, lat2, lon2):
    R = 6371 # Earth radius in kilometers
    phi1, phi2 = np.radians(lat1), np.radians(lat2)
    dphi = np.radians(lat2 - lat1)
    dlamba = np.radians(lon2 - lon1)
    a = np.sin(dphi / 2)**2 + np.cos(phi1) * np.cos(phi2) * np.sin(dlamba / 2)**2
    return 2 * R * np.arctan2(np.sqrt(a), np.sqrt(1 - a))

# Main processing function
def generate_preference_rating(yelp_data):
    # Calculate distance from target location
    yelp_data['distance_km'] = yelp_data.apply(
        lambda row: haversine(target_location[0], target_location[1],
                               row['latitude'], row['longitude']), axis=1
    )

    # Normalize distance to [0, 1] and invert (closer is better)
    scaler = MinMaxScaler()
    yelp_data['location_score'] = 1 - scaler.fit_transform(yelp_data[['distance_km']

```

```

# Normalize business_stars and reviewer_stars to [0, 1]
yelp_data['business_stars_norm'] = yelp_data['business_stars'] / 5
yelp_data['reviewer_stars_norm'] = yelp_data['reviewer_stars'] / 5

# Calculate engagement_score and normalize
yelp_data['engagement_score_raw'] = yelp_data['useful'] + yelp_data['funny'] +
yelp_data['engagement_score'] = scaler.fit_transform(yelp_data[['engagement_sco

# Calculate preference_rating with non-linear transformations
yelp_data['preference_rating'] = (
    weights['location_score'] * apply_non_linear_transformation(yelp_data['loca
    weights['business_stars'] * exponential_boost(yelp_data['business_stars_nor
    weights['reviewer_stars'] * angular_score_adjustment(yelp_data['reviewer_st
    weights['engagement_score'] * np.log1p(yelp_data['engagement_score']))
)

# Normalize the preference_rating to 1-10 scale
rating_scaler = MinMaxScaler(feature_range=(1, 10))
yelp_data['preference_rating'] = rating_scaler.fit_transform(
    yelp_data[['preference_rating']]
).round().astype(int)

# Convert business_id to float
yelp_data['business_id'] = yelp_data['business_id'].apply(business_id_to_float)

return yelp_data

# Data preparation and display
def prepare_final_ranking(yelp_data):
    # Remove duplicates, keeping highest-rated business for each name
    unique_data = yelp_data.sort_values(by='preference_rating', ascending=False).dr

    # Create ranked dataframe
    ranked_yelp = unique_data[['name', 'preference_rating']].sort_values(by='prefer

    return ranked_yelp

# Process the data
yelp_data = generate_preference_rating(yelp_data)

# Prepare and display ranking
ranked_results = prepare_final_ranking(yelp_data)
print(ranked_results)

```

	name	preference_rating
13979	Those Guys Pies	10
17004	Lakes Lounge	10
11268	Yojié Japanese Fondue	10
20834	Sidestreet Pho and Grill	10
12992	Pan Asian	10
...	...	...
9726	Rincon Catracho	3
3742	Sbarro	3
5165	Bogey's Bar & Grill	3
5121	Sushi Culiacan	3
12888	360 Gourmet Burrito	2

[1474 rows x 2 columns]

Using every feature as input to the training process is not a good idea. Some features are not related to business rating, so let's exclude them from our feature set. Let the criteria for a good restaurant be a rating greater than **eight**.

1. Extract the following for the input data `X`:

`business_stars`, `review_count`, `checkins`, `reviewer_stars`, `useful`, `funny`, and `cool`.

2. Store this binary classification variable in `Y`.

```
In [30]: # Initialize X and Y
# ANSWER STARTING HERE

X = yelp_data[['business_stars', 'review_count', 'checkins', 'reviewer_stars', 'useful', 'funny', 'cool']]
Y = (yelp_data['preference_rating'] > 8).astype(int)

print(X.head())
print(Y.head())
# ANSWER ENDS HERE
```

	business_stars	review_count	checkins	reviewer_stars	useful	funny	cool
0	3.5	943	13.0	2.0	1.0	0.0	0
1	3.5	802	7.0	3.0	0.0	0.0	0
2	4.0	1301	62.0	3.0	1.0	0.0	0
3	3.5	5950	673.0	1.0	0.0	0.0	0
4	2.5	159	1.0	3.0	2.0	1.0	2
0	0						
1	0						
2	0						
3	0						
4	0						

Name: preference_rating, dtype: int64

3. Use `train_test_split` from the Scikit-learn module to split the training data using the proposed 80/20 ratio.

```
In [31]: seed = 42 # always use this seed number as your random_state in the whole notebook
test_size=0.2
```

```
# Split your training data and use the seed to get consistent results.
# ANSWER STARTING HERE

X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=test_size, rand

# ANSWER ENDS HERE
print(X_train.shape)
print(X_test.shape)
print(y_train.shape)
print(y_test.shape)
```

```
(17054, 7)
```

```
(4264, 7)
```

```
(17054,)
```

```
(4264,)
```

4. **Normalize** both the training and test sets so that the model is trained on features with a consistent scale. **You can use any scaler from the Scikit-learn module to do this.** If we don't normalize the data, the model may struggle to converge during training because some features could dominate the loss function due to their larger scale. By normalizing the data, we ensure that each feature contributes equally to the learning process, improving the model's ability to find an optimal solution. (eg. [StandardScaler documentation](#))

```
In [32]: #You can add more than 1 scalar. This is the base code.
# ANSWER STARTING HERE
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.fit_transform(X_test)
# ANSWER ENDS HERE
```

Part C (15 Pts Total)

This is going to be the main source of evaluation for the return offer. Be thorough with your model search. You will train a **Logistic Regression Model**, plus 3 different classifiers: **KNN**, **DecisionTree**, **Logistic Regression** (with maximum 1000 iterations) and **Random Forest**.

3.1 Load Models (4 pts)

1. Create a python dictionary named `models` that maps the names of the models to their constructors.

```
In [33]: # Initialize the models
# ANSWER STARTING HERE
models = {
    "Logistic Regression": LogisticRegression(max_iter=1000, random_state=42),
    "K-Nearest Neighbors": KNeighborsClassifier(),
    "Decision Tree": DecisionTreeClassifier(random_state=42),
```

```
"Random Forest": RandomForestClassifier(random_state=42)
}

# ANSWER ENDS HERE
```

3.2 K-fold Cross-Validation (8 pts)

Using the standardized feature sets, do the following:

1. Given five folds and the object with the specified number of splits, perform **k-folds cross-validation** for **each** model using StratifiedKFold.
2. Print the **name** of the model and **cross validation accuracy** (mean and standard deviation) for each.

Cross Validation Documentation

```
In [ ]: # Perform k-fold Cross-Validation for each model
k_folds = 5
skf = StratifiedKFold(n_splits=k_folds, shuffle=True, random_state=seed)

for model_name, model in models.items():
    np.random.seed(42)
    print(f"Model: {model_name}")
    # ANSWER STARTING HERE

    cv_scores = cross_val_score(model, X_train_scaled, y_train, cv=skf, scoring='ac
    print(f"Cross-validated Accuracy: {cv_scores.mean():.4f} (+/- {cv_scores.std():
    # ANSWER ENDS HERE
```

Model: Logistic Regression
Cross-validated Accuracy: 0.9329 (+/- 0.0009)

Model: K-Nearest Neighbors
Cross-validated Accuracy: 0.9299 (+/- 0.0022)

Model: Decision Tree
Cross-validated Accuracy: 0.9643 (+/- 0.0026)

Model: Random Forest
Cross-validated Accuracy: 0.9539 (+/- 0.0026)

3.3 Question: What is the most accurate ML model from above choice? Why? (3 pts)

The most accurate machine learning model is the Decision Tree model, which achieved the highest average accuracy out of all four models. Decision trees can capture complex and non-linear relationships between features. They can learn rules and interactions without requiring feature scaling. It is flexible which is why I think it fit the Yelp data precisely.

3.3 Question: What is the most accurate ML model from above choice? Why? (3 pts)

Part D (12 Pts Total)

We want you to deploy all of the models to production. Evaluate all of the models on the heldout test set.

4.1 Model Evaluation (6 pts)

For each of the models do the following in order to evaluate each of them properly:

- **Train** each of the models
- Print each of the **accuracies** (3 decimal precision) of the models by their **names** in this format: *Accuracy of (model name): (accuracy of the model)*.
- Print a **classification report** with the accuracy, precision, recall and f1-score of each model.

```
In [36]: # Train each model using the training data
for model_name, model in models.items():
    # Set random seed for reproducibility
    np.random.seed(42)

    # ANSWER STARTING HERE
    model.fit(X_train_scaled, y_train)

    # ANSWER ENDS HERE

# Evaluate the performance of each model
for model_name, model in models.items():
    np.random.seed(42)

    # ANSWER STARTING HERE
    y_pred = model.predict(X_test_scaled)
    accuracy = accuracy_score(y_test, y_pred)
    print(f"Accuracy of {model_name}: {accuracy:.3f}") # Your accuracy table header
    print(classification_report(y_test, y_pred)) # Implement your classification re
    # ANSWER ENDS HERE
```

Accuracy of Logistic Regression: 0.931

	precision	recall	f1-score	support
0	0.93	1.00	0.96	3974
1	0.36	0.01	0.03	290
accuracy			0.93	4264
macro avg	0.65	0.51	0.50	4264
weighted avg	0.89	0.93	0.90	4264

Accuracy of K-Nearest Neighbors: 0.923

	precision	recall	f1-score	support
0	0.95	0.97	0.96	3974
1	0.39	0.24	0.30	290
accuracy			0.92	4264
macro avg	0.67	0.61	0.63	4264
weighted avg	0.91	0.92	0.91	4264

Accuracy of Decision Tree: 0.892

	precision	recall	f1-score	support
0	0.94	0.94	0.94	3974
1	0.22	0.23	0.23	290
accuracy			0.89	4264
macro avg	0.58	0.59	0.59	4264
weighted avg	0.89	0.89	0.89	4264

Accuracy of Random Forest: 0.923

	precision	recall	f1-score	support
0	0.95	0.97	0.96	3974
1	0.39	0.25	0.30	290
accuracy			0.92	4264
macro avg	0.67	0.61	0.63	4264
weighted avg	0.91	0.92	0.91	4264

4.2 Model Interpretation (6 pts)

- **Interpret** the results of the classification models (list the models which got the best scores for Accuracy, Precision, Recall and F1 score).
- **Recommend the best model according to your analysis.** Hopefully your best model has over a 90% accuracy as this will guarantee your return offer.

Question: Which model according to your analysis was the best at predicting the Restaurant Preference Rating? Why? DO NOT just say that the accuracy is better than the rest of the models. That is obvious.

Best accuracy: Logistic Regression Best Precision: Tied between Random Forest and K-Nearest Best Recall: Logistic Regression Best F1 Score: Tied between Random Forest and K-Nearest

Even though accuracy is high, class 0 dominates. Even a lazy model could reach 93% accuracy by chance. Saying 0 every time would result in 93% accuracy due to the fact that the 0/1 class split is 93/7.

In logistic regression, we see that its precision for excellent restaurants is .36 with a recall of .01. It is catching almost none of the good restaurants. I think the accuracy statistic is very misleading here. It identified maybe 3 out of 290 good restaurants.

The KNN model has a precision of .39 and a recall of .24. It is finding nearly 25% of all the great restaurants. This is far better in my opinion.

The decision tree outperforms the logistic regression but underperforms compared to the KN models.

The random forest model has a precision of .39 and a recall of .25, which just shyly outperforms the KNN. Its f1-score is also tied with KNN.

I think the best overall model would be the random forest model. It has a very high accuracy and identifies the highest amount of high-rated restaurants. It slightly outperforms KNN, which is in my opinion the second best model.

By apparent accuracy, logistic regression wins. But it fails to detect the restaurants we care about. It only has high accuracy because it spams "0" for like every possibility, which results in 93% due to 0 being correct 93% of the time.

Part E (5 Pts Total)

Seeing all the effort and hard work, it looks as if most people have done well with this project 🎉! With all the input features you have used, there is still no concrete way of figuring out whether a feature is more important than another. We only really used intuition for that. Thus, we shall do some feature engineering to determine this.

5.1 What are your top 4 features used to predict Our restaurant preference rating (5 pts)?

- Use Decision Tree to create a dataframe for **feature importance** named **feature_importance_df**.
- **Extract, Sort, and Print** the **top 4** features from that dataframe.

Here are some references to help you:

[Feature Importance 1](#)

Feature Importance 2

```
In [38]: # Extract the feature importances from the most accurate model
# ANSWER STARTING HERE
rf_model = models["Random Forest"]
importances = rf_model.feature_importances_

# Creating a DataFrame to display feature importances
feature_importance_df = pd.DataFrame({ 'Feature' : X.columns, 'Importance': importances })

# Displaying feature importances
print("Feature Importances:")
print(feature_importance_df[:4])

# ANSWER ENDS HERE
```

Feature Importances:

	Feature	Importance
0	review_count	0.517024
1	checkins	0.189540
2	business_stars	0.091074
3	reviewer_stars	0.076128

Part F (14 Points Total)

Now you'll learn about clustering techniques! You'll perform data preprocessing, apply dimensionality reduction technique, and use K-means clustering to analyze the data. Finally, you'll visualize the results.

We wish to standardize the feature sets in order to make sure that the PCA is not biased by differences in scales within the data. Standardization aids with consistency in the data.

6.1.1 Normalize the data using StandardScaler. (1 pts)

- **Standardize** the features `X` and save it as `X_scaled`. (Include only the feature columns not the label column)

Hint: [documentation](#)

```
In [40]: X = yelp_data[['business_id', 'business_stars', 'review_count', 'checkins', 'review_count', 'reviewer_stars']]

# TODO: Normalize the data using StandardScaler
# ANSWER STARTING HERE
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# ANSWER ENDS HERE
```

There are varying reasons for using a dimensionality reduction step such as PCA prior to data segmentation. By reducing the number of features, we are improving the performance of our algorithm. In addition, by decreasing the number of features the noise is also reduced.

PCA

Principal Component Analysis (PCA) is a dimensionality reduction technique that finds directions (principal components) maximizing variance in a dataset. It's an orthogonal linear transformation mapping data to a new coordinate system [1].

PCA steps for a dataset X with n samples and p features:

1. Center the data.
2. Compute covariance matrix $\Sigma = \frac{1}{n-1} X^T X$.
3. Calculate eigenvectors and eigenvalues of Σ .
4. Sort eigenvectors by descending eigenvalues.
5. Select top k eigenvectors as new basis.

Project original data onto this new basis for transformed data.

PCA is simple and effective for linear relationships but may not suit complex, non-linear structures [2].

For more in-depth reading on PCA:

[1] Scikit-learn PCA documentation: [sklearn.decomposition.PCA](https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html)

[2] Jolliffe, I. T. (2002). Principal Component Analysis, Second Edition. Springer Series in Statistics.

6.1.2 Use a PCA function to return the transformed data. (3 pts)

- Complete the PCA Implementation Function
- Apply Principal Component Analysis to Reduce Dimensionality
- Iteratively adjust the number of components `n_components` to retain at least 80% of the variance
- Save the resulting components for the elements in our data set as `X_pca`

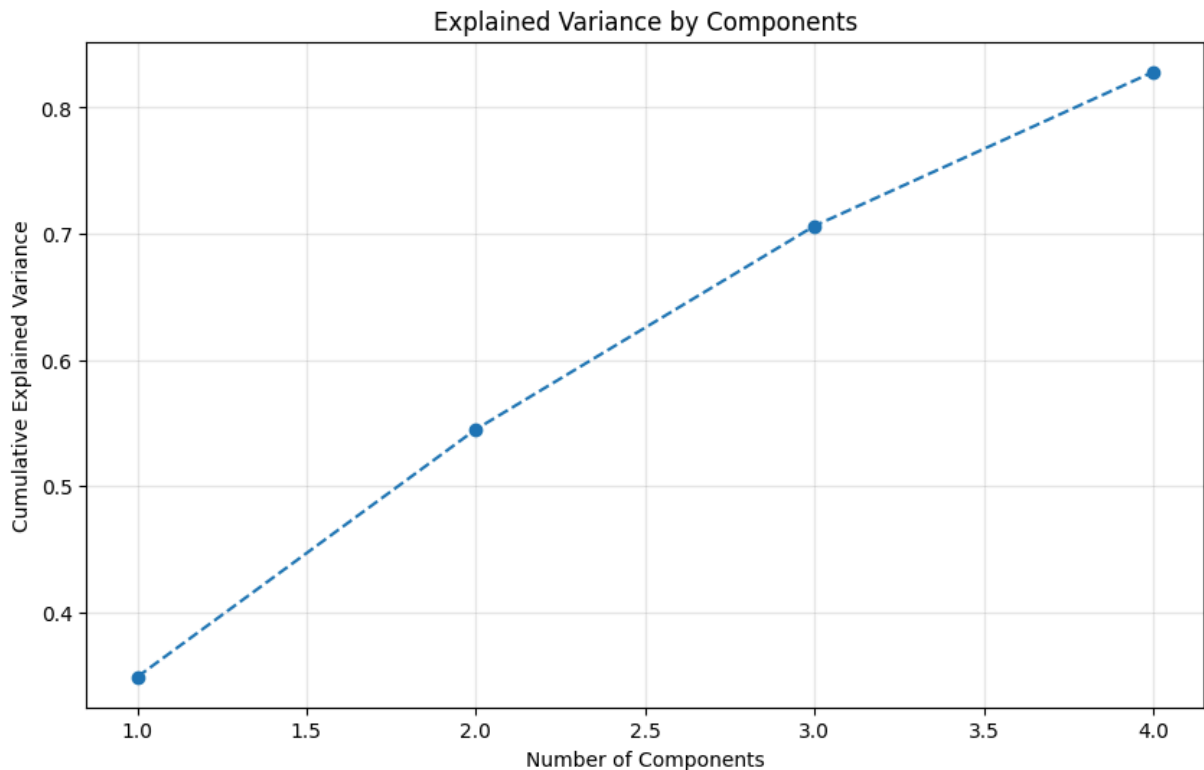
```
In [41]: # TODO: Apply PCA and return the transformed data

def apply_pca(X_scaled, n_components):
    # ANSWER STARTING HERE
    pca = PCA(n_components = n_components, random_state=seed)
    X_pca = pca.fit_transform(X_scaled)
    # ANSWER ENDS HERE
    plt.figure(figsize=(10, 6))
    plt.plot(range(1, n_components + 1), pca.explained_variance_ratio_.cumsum(), marker='o')
    plt.title('Explained Variance by Components')
    plt.xlabel('Number of Components')
    plt.ylabel('Cumulative Explained Variance')
    plt.grid(True, alpha=0.3)
    plt.show()
```

```

return pca, X_pca
# Apply PCA
n_components = 4
pca, X_pca = apply_pca(X_scaled, n_components)
print(X_pca.shape)
print(pca.explained_variance_ratio_)

```



```

(21318, 4)
[0.34858963 0.19627698 0.16126539 0.12203854]

```

6.2.1 Elbow Method (2.5 pts)

In K-Means clustering, we start by randomly initializing k clusters and iteratively adjusting these clusters until they stabilize at an equilibrium point. However, before we can do this, we need to decide how many clusters (k) we should use.

In this part, you will determine the optimal number of clusters for a given dataset using the **Elbow Method**. The goal is to identify the point where adding more clusters results in minimal improvement in clustering quality, known as the "elbow point."

The *distortion* for a given number of clusters k can be represented as:

$$D(k) = \frac{1}{N} \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$

Where:

- $D(k)$ is the distortion for k clusters
- k is the number of clusters

- C_i represents the i -th cluster
- x is each data point in the cluster
- μ_i is the centroid (mean) of the i -th cluster
- $\| \cdot \|$ denotes the Euclidean distance

Task A: Implement your **own distortion function** to determine the different numbers of clusters, ranging from 1 to 9.

- Store your mean distortion values in:
 - A list named **distortions** (one value per k)
 - A dictionary named **mapping** with keys as k values and values as their corresponding mean distortions
- Create an elbow plot with k on the x-axis and mean distortion on the y-axis
- Visually identify the elbow point (the optimal value of k)

Use `X_pca` that we calculated in the previous parts for this part. (**WARNING: You can't use the `KMeans.inertia_`**)

```
In [42]: distortions = []
K = range(1, 10)
mapping = {}
for k in K:
    # ANSWER STARTING HERE
    means = KMeans(n_clusters=k, random_state=seed)
    means.fit(X_pca)

    distances = np.min(cdist(X_pca, means.cluster_centers_, 'euclidean'), axis=1)
    distortion = np.mean(distances ** 2)

    distortions.append(distortion)
    mapping[k] = distortion

    # ANSWER ENDS HERE

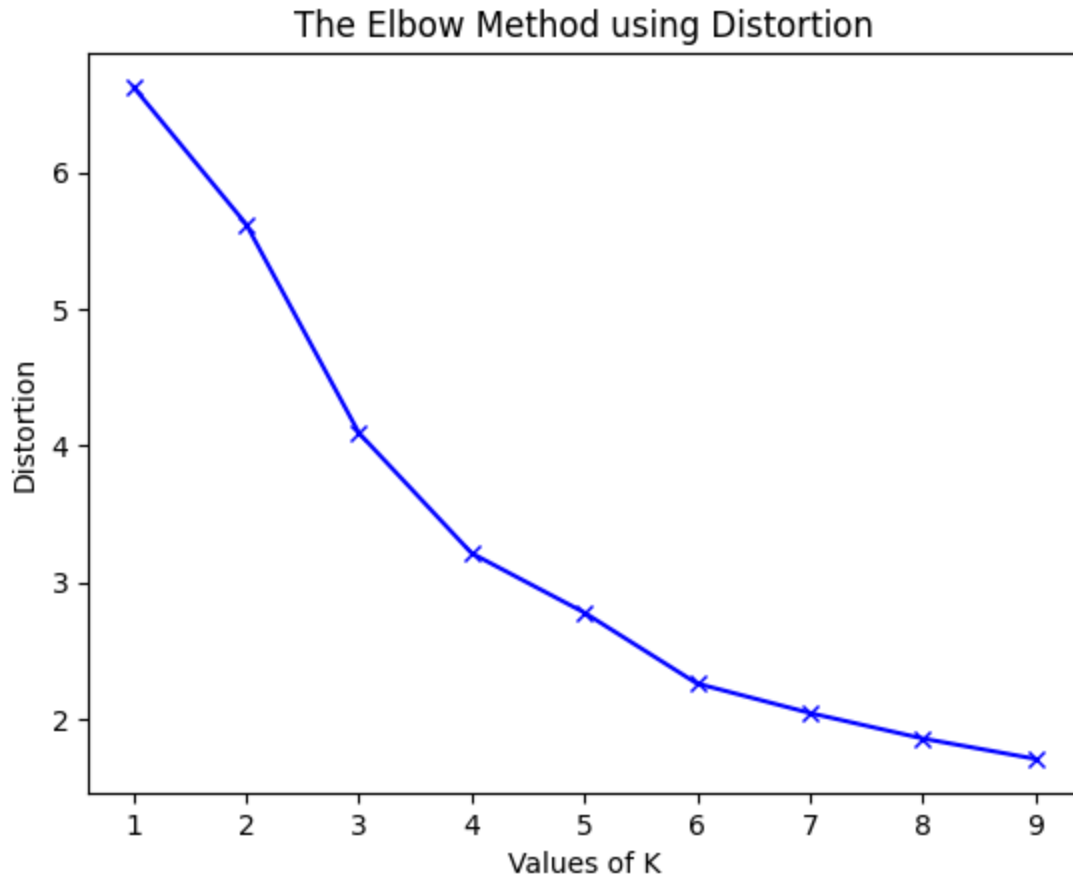
for key, val in mapping.items():
    print(f'{key} : {val}')

plt.plot(K, distortions, 'bx-')
plt.xlabel('Values of K')
plt.ylabel('Distortion')
plt.title('The Elbow Method using Distortion')
plt.show()
```

```

1 : 6.625364385427708
2 : 5.621408448719423
3 : 4.0950293076122675
4 : 3.214770463491667
5 : 2.7808655934855446
6 : 2.2616488292651207
7 : 2.0465346184567155
8 : 1.860069897820657
9 : 1.7109364333757937

```



6.2.2 KElbowVisualizer (2.5 pts)

Task B: Use the `KElbowVisualizer` from the `yellowbrick` library to verify the conclusion above. See [Elbow Method](#), and [here](#) to install yellowbrick.

```

In [43]: from sklearn import datasets
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt
from yellowbrick.cluster import SilhouetteVisualizer
from yellowbrick.cluster import KElbowVisualizer
import yellowbrick.cluster.elbow as elbow_module
from yellowbrick.cluster.elbow import distortion_score

def mean_distortion_score(X, labels, metric="euclidean"):
    """Compute the mean distortion (average squared distance to centroids)"""
    total_distortion = distortion_score(X, labels, metric)
    return total_distortion / X.shape[0]

```

```

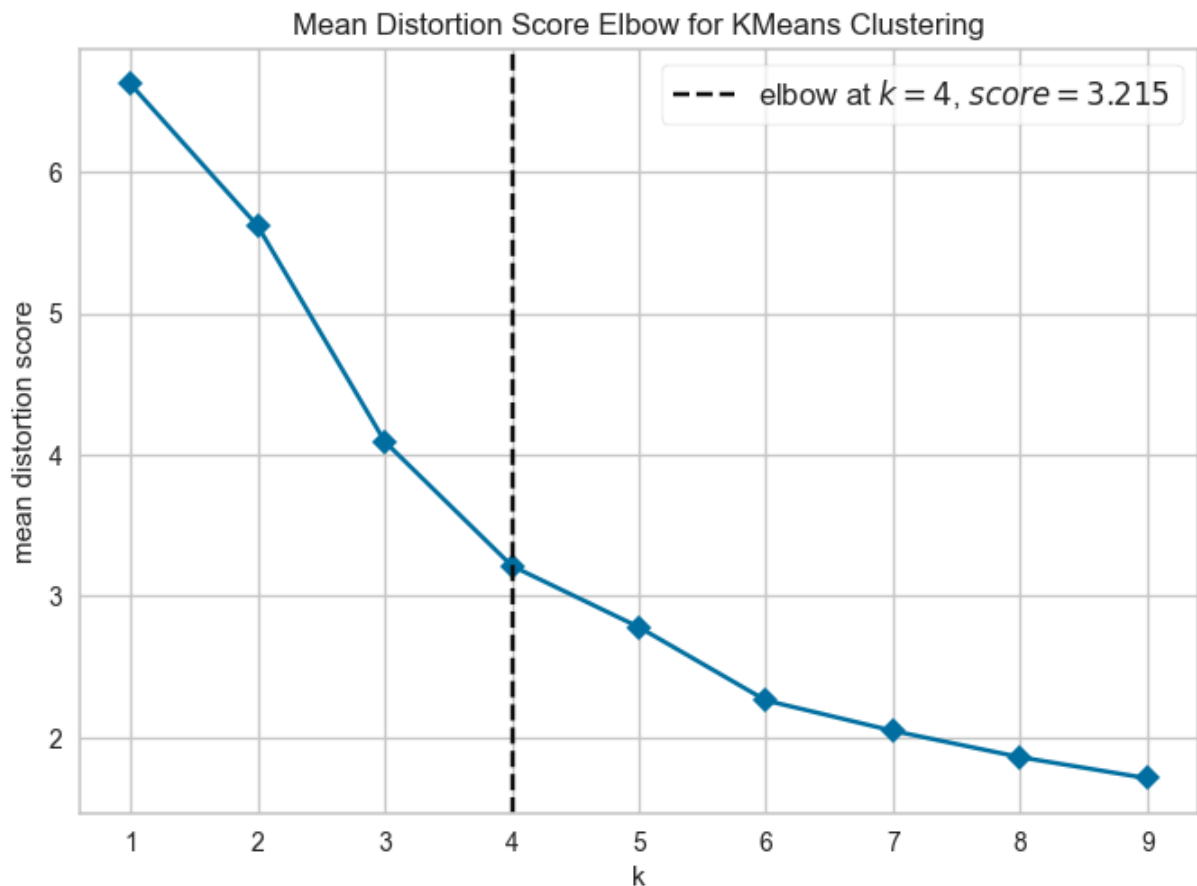
elbow_module.KELBOW_SCOREMAP['distortion'] = mean_distortion_score

# ANSWER STARTING HERE
# Load the dataset
X_yellowbrick = X_pca

# Instantiate the clustering model and visualizer
model = KMeans(random_state=seed)
visualizer = KElbowVisualizer(model, k=(1,10), metric='distortion', timings=False)
visualizer.fit(X_yellowbrick)
visualizer.show()

# ANSWER ENDS HERE

```



Out[43]: <Axes: title={'center': 'Mean Distortion Score Elbow for KMeans Clustering'}, xlabel='k', ylabel='mean distortion score'>

To determine the optimal number of clusters, we have to select the value of k at the “elbow”, i.e. the point after which the distortion starts decreasing in a linear fashion. Thus for the given data, we conclude that the optimal number of clusters for the data is 4.

In [44]: k_cluster = 4

6.2.3 Question: According to the plots above, why do you think we chose $K=4$? (1 pts)
What distortion value do the last two plots show, and what does it indicate about the clustering? (1 pts)

ANSWER STARTING HERE

I think we chose $K=4$ because in both the manual elbow method and keblow method, a clear elbow was shown at this point. Both show a distortion value of around 3.25 at $K=4$. This shows how good the clustering fit is. The flattening shows us when adding clusters stops helping.

K-Means Clustering

K-Means is a popular unsupervised learning algorithm used for clustering, where the goal is to partition data into K clusters based on feature similarity. The algorithm works by iteratively assigning data points to the nearest centroid (cluster center) and updating centroids to minimize the variance within each cluster. It starts by randomly initializing centroids and repeats the process until the centroids stabilize (convergence). K-Means is efficient for large datasets but sensitive to the choice of K and initial centroids. In your homework, you'll implement K-Means to explore how well it groups your data based on similarity!

6.4 Apply K-means clustering and return the cluster labels (3 pts)

```
In [46]: # TODO: Apply K-means clustering and return the cluster labels
def apply_kmeans(X, n_clusters, random_state=seed):
    # ANSWER STARTING HERE
    kmeans = KMeans(n_clusters=n_clusters, random_state=random_state)
    kmeans.fit(X)
    labels = kmeans.labels_
    # ANSWER ENDS HERE
    return labels

# Apply K-means to both PCA results
pca_labels = apply_kmeans(X_pca, n_clusters = k_cluster)
pca_labels.shape
```

Out[46]: (21318,)

Visualize your clustering!

```
In [47]: ## Step 6: Visualization

def visualize_3d_plotly_interactive(X_pca, pca_labels):
    """Fully interactive 3D visualization with Plotly using pre-computed labels"""

    k_cluster = len(np.unique(pca_labels))

    # Create the plot
    fig = go.Figure()

    # Add cluster points
    colors = px.colors.qualitative.Set1[:k_cluster]
    for i in range(k_cluster):
        cluster_mask = pca_labels == i
```

```

fig.add_trace(go.Scatter3d(
    x=X_pca[cluster_mask, 0],
    y=X_pca[cluster_mask, 1],
    z=X_pca[cluster_mask, 2],
    mode='markers',
    marker=dict(
        size=5,
        color=colors[i],
        opacity=0.7
    ),
    name=f'Cluster {i}',
    text=[f'Cluster {i} Point {j}' for j in np.where(cluster_mask)[0]],
    hovertemplate='**%{text}**<br>' +
        'PC1: %{x:.2f}<br>' +
        'PC2: %{y:.2f}<br>' +
        'PC3: %{z:.2f}<br>' +
        '<extra></extra>'
))

# Calculate centroids from the labeled data
centroids = []
for i in range(k_cluster):
    cluster_points = X_pca[pca_labels == i]
    centroid = np.mean(cluster_points, axis=0)
    centroids.append(centroid)

centers_3d = np.array(centroids)[: , :3]

# Add centroids
fig.add_trace(go.Scatter3d(
    x=centers_3d[:, 0],
    y=centers_3d[:, 1],
    z=centers_3d[:, 2],
    mode='markers',
    marker=dict(
        size=8,
        color='red',
        symbol='x',
        line=dict(width=2, color='darkred')
    ),
    name='Centroids',
    text=[f'Centroid {i}' for i in range(k_cluster)],
    hovertemplate='**%{text}**<br>' +
        'PC1: %{x:.2f}<br>' +
        'PC2: %{y:.2f}<br>' +
        'PC3: %{z:.2f}<br>' +
        '<extra></extra>'
))

# Update Layout
fig.update_layout(
    title='Interactive 3D Cluster Visualization',
    scene=dict(
        xaxis_title='Principal Component 1',
        yaxis_title='Principal Component 2',
        zaxis_title='Principal Component 3',

```

```
        camera=dict(  
            eye=dict(x=1.2, y=1.2, z=1.2)  
        )  
    ),  
    width=900,  
    height=700,  
    showlegend=True  
)  
  
fig.show()  
  
return pca_labels  
  
# Usage with your existing pca_labels  
cluster_labels = visualize_3d_plotly_interactive(X_pca, pca_labels)
```

Your analysis here

You have been promoted!!

You are now one of the consulting company's elite employees! 🥳🥳🥳🥳🥳



Part G (2 points)

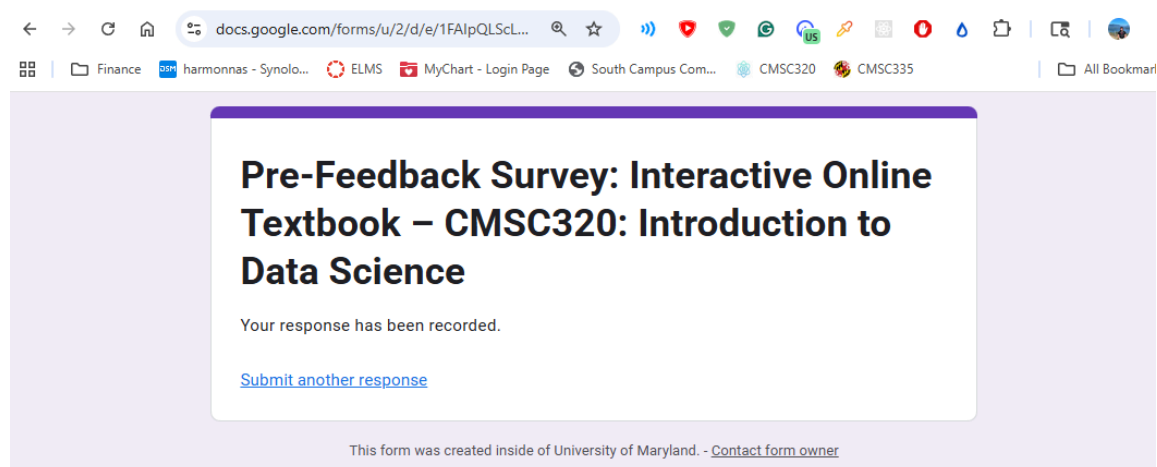
We have shared an online interactive textbook for CMSC320: Introduction to Data Science. The textbook is still in progress, and more chapters will be published gradually throughout the semester.

This short pre-feedback survey will help us understand your experience, preferences, and expectations regarding the textbook format before you start using it.

Here is the link to the survey: <https://forms.gle/n3Nbmeosoos7tjFa9>

Here is the link to the textbook: <https://ffalam.github.io/CMSC320TextBook/index.html>

Note: After completing the survey, please take a screenshot of your submission and submit it as proof of participation. This is a compulsory question. If you have already completed the survey earlier, you do not need to do it again, but you must submit the previous screenshot to receive the points.



Extra Credit 🌟 (5 points total)

The matplotlib and seaborn libraries are great tools for visualizing data.

They are used to create legible and complex graphs that are essential in data interpretation and analysis.

7.1 Visualize the distribution of each feature in the original dataset for different rating classes (2 pts)

- **Plot** out histograms for each feature. You can use a for loop.
- Label the **title** of each histogram "Distribution of (Feature Name) by Rating"
- Label the **axes** appropriately and display each histogram.
- Display the **y-axis** using a **logarithmic** scale.

```
In [48]: histogram_pd = X
         histogram_pd.head()
```

```

rating_classes = yelp_data['preference_rating']

# Create subplots
n_features = histogram_pd.shape[1] - 1
n_cols = 3
n_rows = int(np.ceil(n_features / n_cols))

fig, axes = plt.subplots(n_rows, n_cols, figsize=(15, 12))
axes = axes.flatten() # Make it easier to index

# ANSWER STARTING HERE

# Plot histogram for each feature
for i, feature in enumerate(histogram_pd.columns[:-1]):
    ax = axes[i]

    # Get unique rating classes
    classes = sorted(rating_classes.unique())

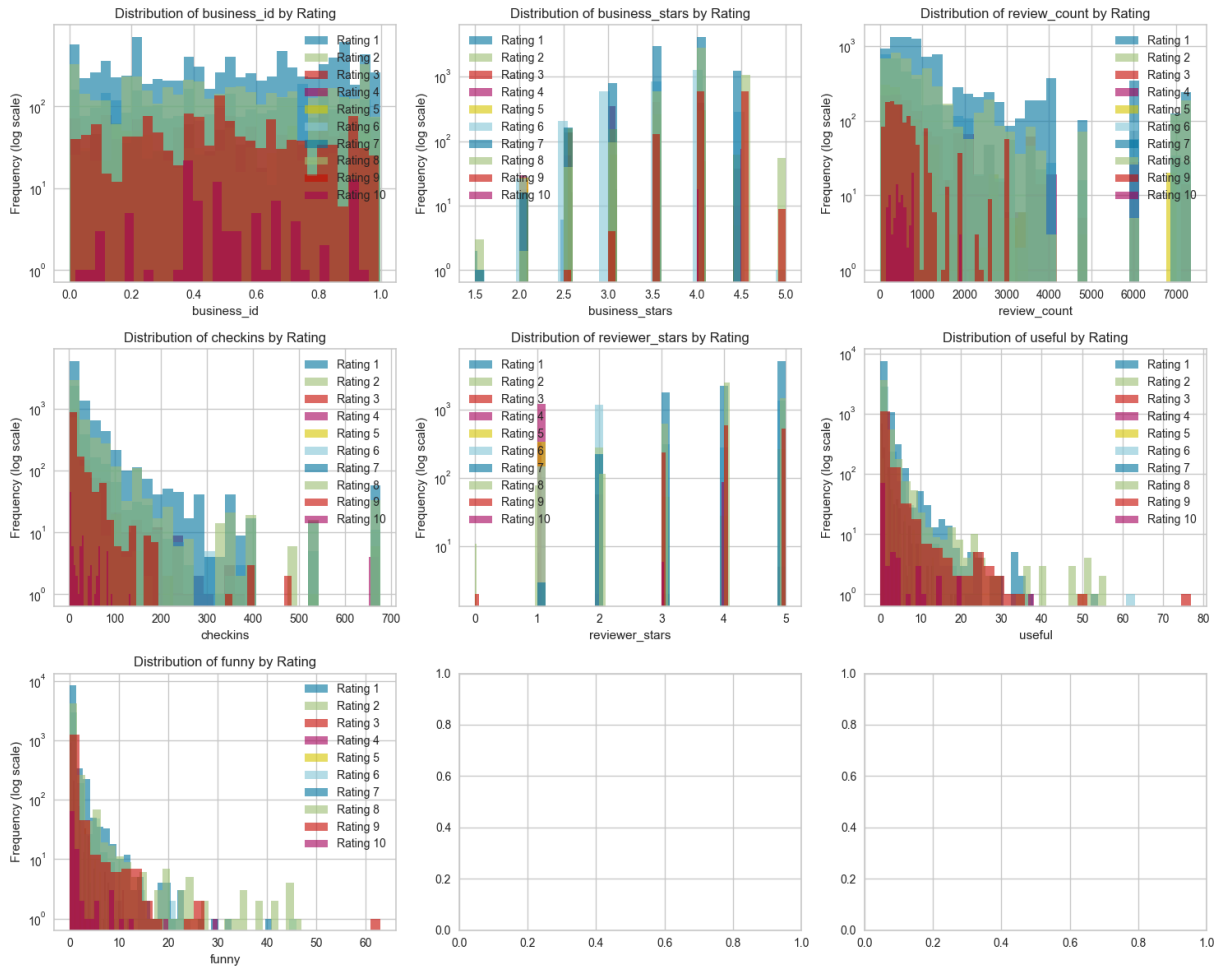
    # Plot histogram for each rating class
    for rating in classes:
        ax.hist(histogram_pd[rating_classes == rating][feature], bins=30, alpha=.6,

    # Set logarithmic scale for y-axis
    ax.set_yscale('log')
    # Set labels and title
    ax.set_xlabel(feature)
    ax.set_ylabel('Frequency (log scale)')
    ax.set_title(f'Distribution of {feature} by Rating')
    ax.legend(fontsize='small')

# ANSWER ENDS HERE

plt.tight_layout()
plt.show()

```



7.2 Explore correlations between features and Rating ratings (1 pts)

- Create a **correlation matrix** based on the correlations between features and rating.
- **Display** the matrix.

In [49]: *# Correlation Matrix*

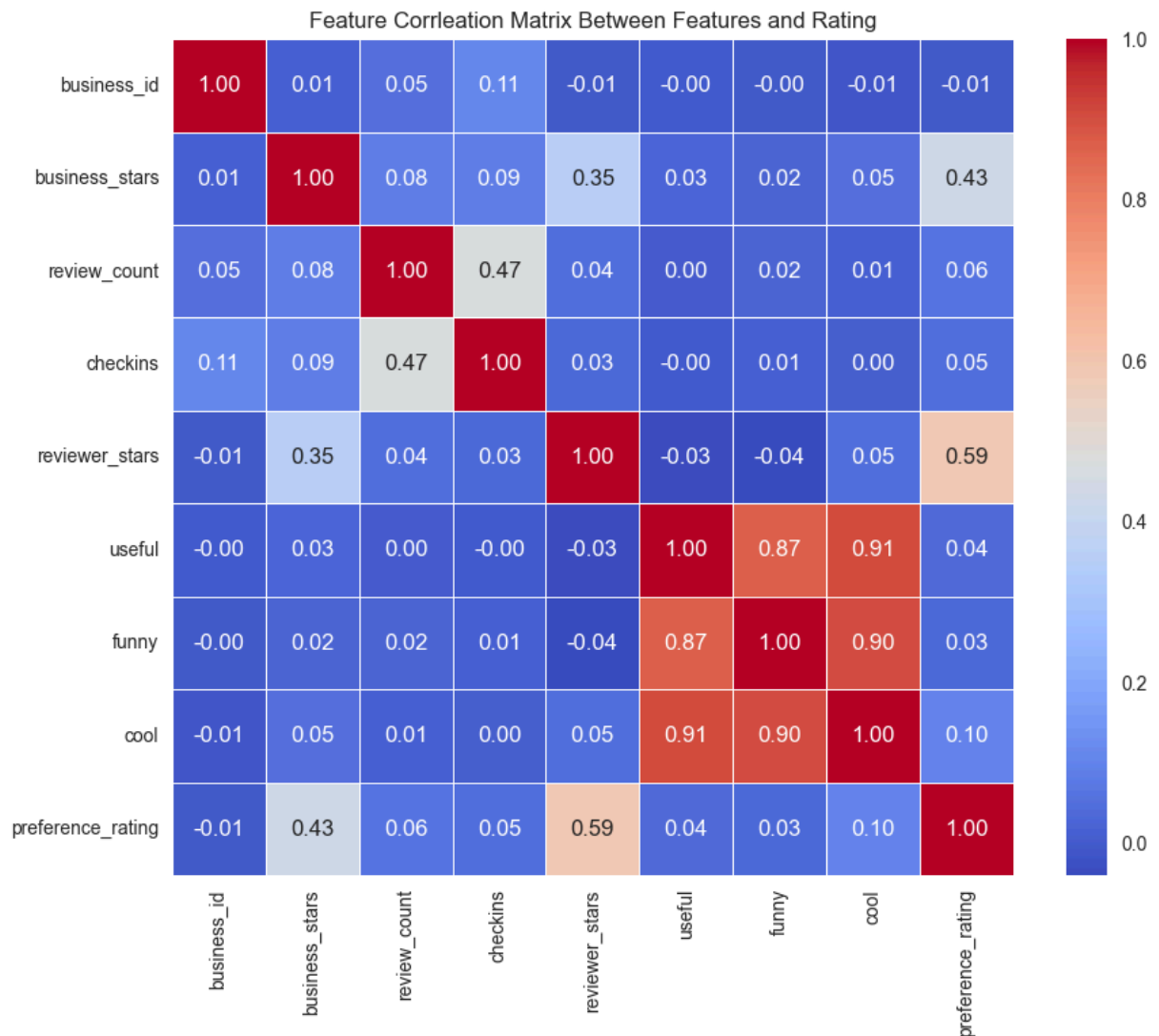
ANSWER STARTING HERE

```
corr_df = X.copy()
corr_df['preference_rating'] = yelp_data['preference_rating']

corr_matrix = corr_df.corr()

plt.figure(figsize=(10,8))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5)
plt.title('Feature Correlation Matrix Between Features and Rating')
plt.show()
```

ANSWER ENDS HERE



7.3 Question: What have you learned from this correlation map? Why? (2 pts)

ANSWER RIGHT HERE!

I learned that reviewer stars and business stars are the strongest predictors of a restaurant's preference rating. They have the values closest to 1 on the graph. This indicates that user perception and satisfaction affect preference rating way more than the amount of checkins or review count. It seems individual user satisfaction matters more than popularity or total checkins.